DIGITAL COMMUNICATIONS

Digital Communications

A waveform becomes bits. A channel carries the bits through noise. A receiver decides what was sent and measures the error.

Covers
Chapters 1 to 6, and appendices A and B

Level
Undergraduate

Assumed background
Fourier analysis, probability, random processes

Course conventions

Read the plain explanation before the mathematics. Worked examples use five headings: Given, Find, Method, Solution, and Check.

Noise is white and Gaussian with **two-sided** power spectral density $N_0/2$ watts per hertz. The Gaussian tail is $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{\infty} e^{-t^2/2}\, dt = \frac{1}{2}\,\mathrm{erfc}(x/\sqrt{2})$. Energy and power use $R = 1\ \Omega$.

A **PS** marker points to the related section of Proakis and Salehi, *Fundamentals of Communication Systems*, second edition. The textbook chapter numbers differ from the course chapter numbers.

CHAPTER 1

The transition from analog to digital

A continuous waveform becomes a bit stream in three steps. Sampling makes time discrete. Quantization makes amplitude discrete. Encoding replaces each quantized amplitude with bits. Sampling can be reversed. Quantization cannot.

1.1 Impulse-train sampling

Sampling every T_s seconds is multiplication by a train of impulses. Write $f_s = 1/T_s$ for the sampling frequency.

> THE IDEAL SAMPLED SIGNAL
>
> $$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$
>
> $$g_\delta(t) = g(t)\,p(t) = \sum_{n=-\infty}^{\infty} g(nT_s)\,\delta(t - nT_s)$$
>
> The second line uses the sampling property $g(t)\delta(t-t_0) = g(t_0)\delta(t-t_0)$, which turns the product under the sum into a number.

The signal g_δ is continuous in time, not a sequence of numbers. Therefore, it has a Fourier transform. A product in time becomes a convolution in frequency:

$$P(f) = \frac{1}{T_s}\sum_{n=-\infty}^{\infty} \delta(f - nf_s) = f_s \sum_{n=-\infty}^{\infty} \delta(f - nf_s)$$

The Fourier-series coefficient of $p(t)$ is the same for every harmonic: $a_k = \frac{1}{T_s}\int_{-T_s/2}^{T_s/2} \delta(t) e^{-j2\pi k f_0 t}\,dt = \frac{1}{T_s}$. Now convolve G with the impulse train:

$$G_\delta(f) = G(f) * P(f) = \frac{1}{T_s}\sum_{n=-\infty}^{\infty} G(f) * \delta(f - nf_s)$$

A shifted impulse shifts the function during convolution. Write the convolution integral and use the even property $\delta(-u) = \delta(u)$:

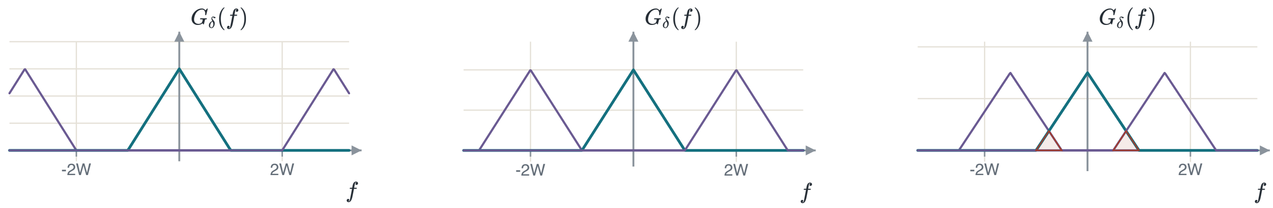
$$\begin{aligned}
G(f) * \delta(f - nf_s) &= \int G(\theta)\,\delta(f - nf_s - \theta)\,d\theta \\
&= \int G(\theta)\,\delta\big(-[\theta - (f - nf_s)]\big)\,d\theta \\
&= \int G(\theta)\,\delta\big(\theta - (f - nf_s)\big)\,d\theta \\
&= G(f - nf_s)
\end{aligned}$$

The last line uses the sifting property. Each impulse in $P(f)$ places one copy of G at a multiple of f_s.

SAMPLING REPLICATES THE SPECTRUM

$$G_\delta(f) = f_s \sum_{n=-\infty}^{\infty} G(f - nf_s)$$

Sampling copies the spectrum to every multiple of f_s and scales it by f_s. Nothing is lost provided the copies do not overlap.



$f_s > 2W$: a gap between the replicas. $f_s = 2W$: they touch and do not overlap. $f_s < 2W$: they overlap, and the overlap cannot be undone.

ALIASING

When the replicas overlap, a high message frequency combines with a low frequency from a replica. No filter can separate these components. The samples no longer determine the original signal. A real signal is not strictly bandlimited, so a **guard band** f_g separates the message and the first replica. Then $f_s = 2W + f_g$. An anti-aliasing filter removes frequencies above W before sampling.

1.2 The sampling theorem and reconstruction

SAMPLING THEOREM

If $G(f) = 0$ for $|f| \geq W$ and $f_s \geq 2W$, then $g(t)$ is determined completely by its samples $g(nT_s)$ and can be recovered from them exactly. If $f_s < 2W$, aliasing occurs and the recovery fails. The rate $2W$ is the **Nyquist rate** and $T_s = 1/(2W)$ the **Nyquist interval**.

An ideal lowpass filter keeps the copy at the origin and rejects the other copies. At $f_s = 2W$:

THE RECONSTRUCTION FILTER AND ITS IMPULSE RESPONSE

$$H_{\mathrm{LPF}}(f) = \begin{cases} \dfrac{1}{2W}, & |f| \leq W \\ 0, & \text{otherwise} \end{cases}$$

$$h_{\mathrm{LPF}}(t) = \int_{-W}^{W} \frac{1}{2W} e^{j2\pi ft}\, df = \frac{\sin(2\pi Wt)}{2\pi Wt} = \mathrm{sinc}(2Wt)$$

The course uses $\mathrm{sinc}(x) = \sin(\pi x)/(\pi x)$. This function is one at zero and zero at every other integer.

RECONSTRUCTION-FILTER GAIN

Sampling scales the spectrum by $f_s = 2W$. The filter must remove this scale factor. A unit-gain filter returns a signal that is $2W$ times too large.

Filtering in frequency is convolution in time. Therefore, convolve the impulse train with h_{LPF}. Use τ as the integration variable:

THE CONVOLUTION INTEGRAL, AND THE SIFTING STEP

$$g_r(t) = \int_{-\infty}^{\infty} \overbrace{\sum_{n=-\infty}^{\infty} g(nT_s)\,\delta(\tau - nT_s)}^{=g_\delta(\tau)}\,\mathrm{sinc}\big(2W(t-\tau)\big)\,d\tau$$

$$g_r(t) = \sum_{n=-\infty}^{\infty} g(nT_s) \underbrace{\int_{-\infty}^{\infty} \mathrm{sinc}\big(2W(t-\tau)\big)\,\delta(\tau - nT_s)\,d\tau}_{=\mathrm{sinc}\left(2W(t-nT_s)\right)}$$

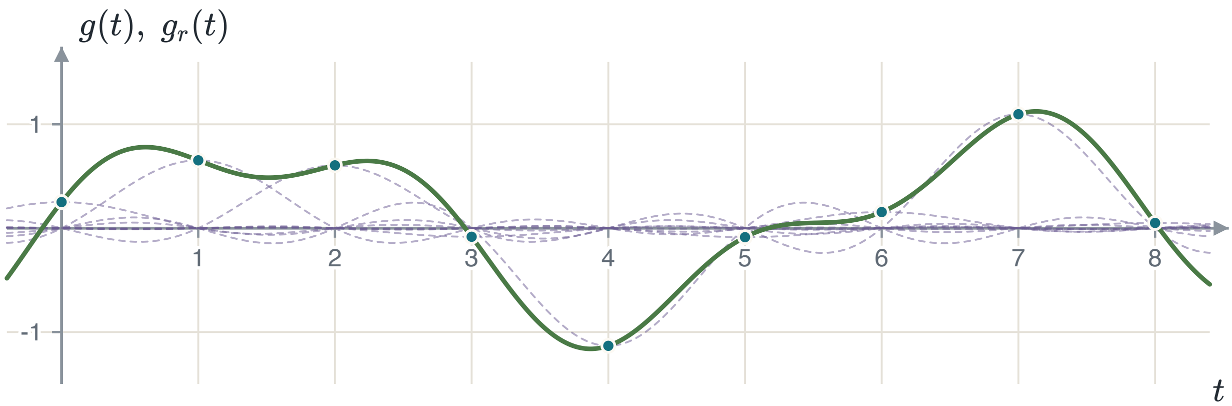
The sum does not depend on τ, so it moves outside the integral. The sifting property then replaces τ with nT_s.

INTERPOLATION

$$g_r(t) = \sum_{n=-\infty}^{\infty} g(nT_s)\,\mathrm{sinc}\big(2W(t - nT_s)\big)$$

$$g_r(t) = \sum_{n=-\infty}^{\infty} g\left(\frac{n}{2W}\right)\mathrm{sinc}(2Wt - n) \quad \text{at } f_s = 2W$$

At $t = kT_s$, every term is zero except the term with $n = k$. Therefore, $g_r(kT_s) = g(kT_s)$. The same sum reconstructs the signal between the sample times.



Each sample contributes one shifted sinc scaled by its own value. Their sum, drawn heavy, passes through every sample because each sinc is zero at all the other sampling instants.

EXAMPLE 1.1 — THREE SAMPLING RATES

GIVEN A signal $x(t)$ bandlimited to $W = 40$ kHz.

FIND Find (a) the Nyquist rate, (b) the rate with a 10 kHz guard band, and (c) the Nyquist rate of $y(t) = x(t)\cos(80000\pi t)$.

METHOD Read (a) and (b) off the geometry of the replicas. For (c) find the bandwidth of y first: the rate follows from that, not from the bandwidth of x.

SOLUTION (a) $f_s = 2W = 80$ kHz. (b) $f_s = 2W + f_g = 90$ kHz. (c) The carrier is at $f_c = 40$ kHz, and multiplication shifts the spectrum both ways and halves it: $Y(f) = \frac{1}{2}X(f - 40\text{k}) + \frac{1}{2}X(f + 40\text{k})$. The result occupies $|f| < 80$ kHz, so $f_s = 160$ kHz.

CHECK The highest frequency of y is $f_c + W = 80$ kHz. Therefore, its Nyquist rate is 160 kHz. The rate follows the highest frequency, not the message bandwidth.

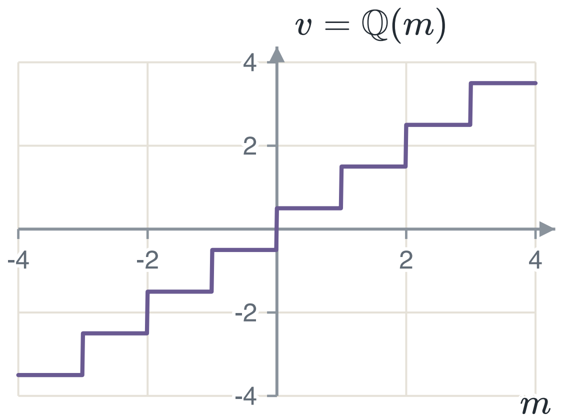
COMMON ERROR IN PART (C)

The rate 80 kHz applies to x, not to y. Modulation moves the highest frequency to 80 kHz. Therefore, y needs a sampling rate of 160 kHz.

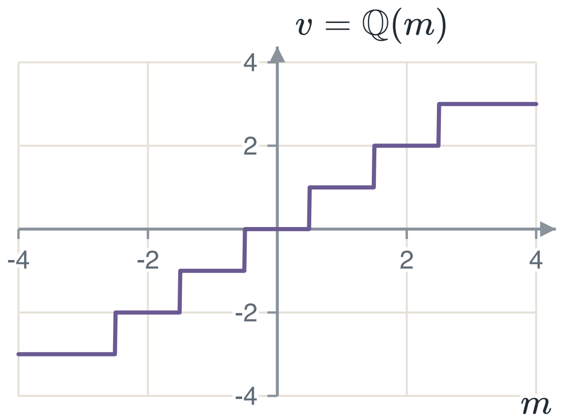
1.3 Quantization

Quantization replaces a sample amplitude with one of L **representation levels**. A **uniform** quantizer uses the same spacing Δ between consecutive levels.

A **mid-rise** quantizer has a decision boundary at zero, so it has no zero output level. A **mid-tread** quantizer has an output level at zero. Thus, a mid-tread quantizer reproduces a small input as zero.



Mid-rise, $L = 8$, $\Delta = 1$. A boundary at the origin.



Mid-tread, $L = 8$, $\Delta = 1$. A level at the origin.

A quantizer divides the input range into L regions $\mathcal{J}_k$. Each region has one output value v_k. Two conditions define an optimal quantizer:

1. Each boundary is the **midpoint** of its two adjacent levels: $m_k = \frac{1}{2}(v_{k-1} + v_k)$. This sends each input to the nearer level.
2. Each level is the **centroid** of its region: $v_k = E[M \mid M \in \mathcal{J}_k]$. This minimizes the mean-square error in that region.

The two conditions depend on each other, so apply them in turn. A uniform quantizer satisfies both conditions for a uniform input.

1.4 Quantization noise and the signal-to-noise ratio

The quantization error is $Q = M - \mathbb{Q}(M)$. Nearest-level quantization keeps this error between $-\Delta/2$ and $\Delta/2$.

If Δ is small, the density of M is nearly constant across one region. The error can then be modeled as uniform.

THE UNIFORM ERROR MODEL

$$f_Q(q) = \frac{1}{\Delta}, \qquad -\frac{\Delta}{2} \le q \le \frac{\Delta}{2}$$

$$E[Q^2] = \int_{-\Delta/2}^{\Delta/2} q^2 \frac{1}{\Delta}\, dq = \frac{\Delta^2}{12}$$

For a zero-mean error, this mean-square value is also the variance and the quantization-noise power.

Substitute $\Delta = 2m_{\max}/L$ and $L = 2^R$. Then divide the signal power by the error power to obtain the SQNR.

SIGNAL-TO-QUANTIZATION-NOISE RATIO

$$E[Q^2] = \frac{1}{12}\left(\frac{2m_{\max}}{L}\right)^2 = \frac{m_{\max}^2}{3 \cdot 2^{2R}}$$

$$\mathrm{SQNR} = \frac{P_M}{E[Q^2]} = \frac{3P_M}{m_{\max}^2} 2^{2R}$$

$$\mathrm{SQNR}\ [\mathrm{dB}] = \underbrace{10 \log_{10} \frac{3P_M}{m_{\max}^2}}_{\alpha} +\ 6.02R$$

Each extra bit adds $20 \log_{10} 2 = 6.02$ dB. Doubling the level count halves the step size and divides the error power by four.

INPUT-RANGE USE

The term α depends on the signal power and peak amplitude. A full-scale sinusoid gives $\alpha \approx 1.76$ dB. A sinusoid that reaches one quarter of the range loses 12.04 dB.

EXAMPLE 1.2 — A FULL-SCALE SINUSOID

GIVEN $m(t) = 5 \cos t$ through a uniform quantizer spanning its full range.

FIND The step size and the SQNR at $R = 3$ and $R = 4$ bits per sample.

METHOD Average power from Parseval, step size from $\Delta = 2m_{\max}/L$, then $\alpha + 6.02R$.

SOLUTION The coefficients are $a_{\pm 1} = 5/2$, so $P_M = 2(5/2)^2 = 12.5$ and $m_{\max} = 5$. At $R = 3$: $\Delta = 2(5)/8 = 1.25$ V and $\mathrm{SQNR} = 1.76 + 18.06 = 19.82$ dB. At $R = 4$: $\Delta = 0.625$ V and $\mathrm{SQNR} = 1.76 + 24.08 = 25.84$ dB.

CHECK The two differ by 6.02 dB, one bit. Independently, $E[Q^2] = \Delta^2/12 = 0.1302$ and $10 \log_{10}(12.5/0.1302) = 19.82$ dB.

MODEL LIMIT

The measured results are 19.09 dB and 25.31 dB. They are below the model by 0.7 dB and 0.5 dB. A sinusoid does not produce a uniform error at low resolution. The gap decreases as R increases.

EXAMPLE 1.3 — WHEN THE MODEL BREAKS

GIVEN A zero-mean stationary Gaussian source has $S_X(f) = 2$ for $|f| < 100$ Hz. Its samples enter a five-level quantizer with outputs $-30, -10, 0, 10, 30$ and boundaries at $-40, -20, 20, 40$.

FIND The SQNR of the scheme.

METHOD The signal power is the area under the spectral density. The noise power must be integrated region by region, because this quantizer is neither uniform nor fine.

SOLUTION $P_X = \int_{-100}^{100} 2\, df = 400$, and since the mean is zero this is also σ_X^2. Splitting $P_Q = \int (x - \mathbb{Q}(x))^2 f_X(x)\, dx$ at the four boundaries gives $7.98 + 46.36 + 79.50 + 46.36 + 7.98 = 188.18$, so $\mathrm{SQNR} = 10 \log_{10}(400/188.18) = 3.27$ dB.

CHECK The central region alone contributes 79.50, and it holds the 68% of the mass with $|X| < 20$ and an error of up to 20. That is where a five-level quantizer spends its error, and it is why the answer is a few decibels rather than a few tens.

MODEL LIMIT

This quantizer is coarse, and its outer regions are unbounded. A sample at $x = 120$ has an error of 90. The formula $\Delta^2/12$ predicts 10.8 dB, but direct integration gives 3.27 dB.

1.5 Non-uniform quantization

Speech contains small amplitudes more often than large amplitudes. A uniform quantizer gives every amplitude the same step size. Thus, its relative error is larger for small amplitudes.

A non-uniform quantizer uses narrow regions where the probability density is high. It uses wide regions where the density is low. This arrangement reduces distortion without adding levels.

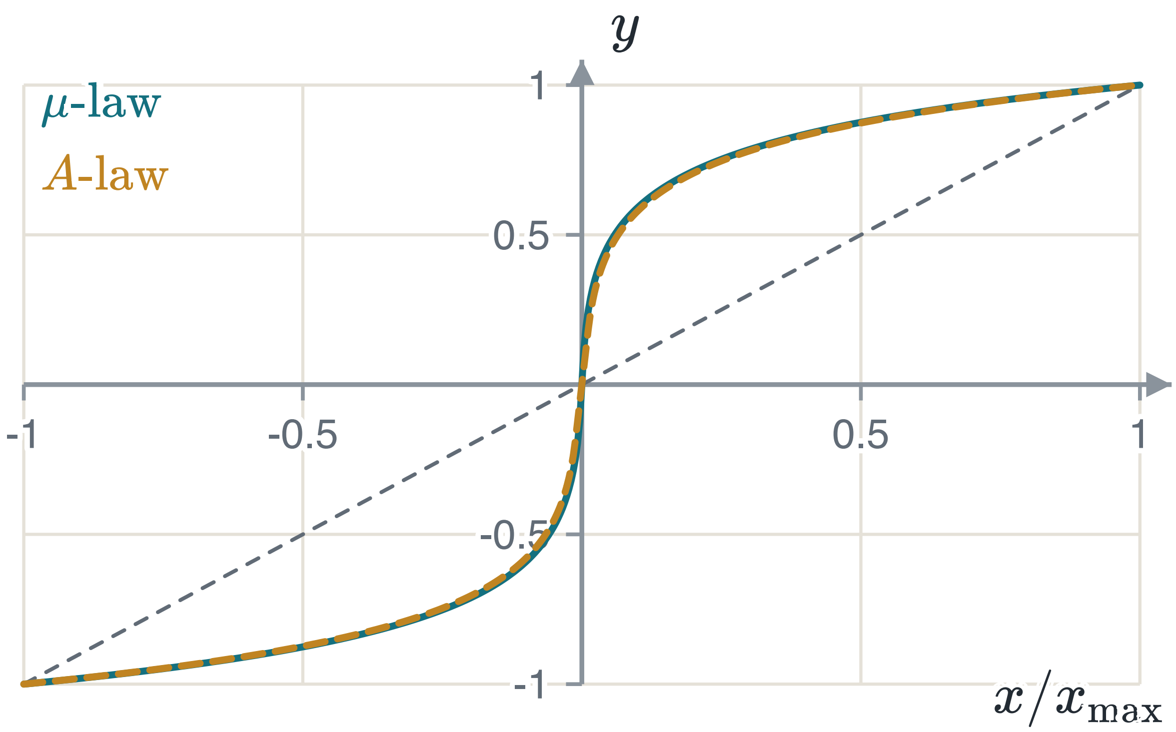
A **compander** first compresses the signal with a memoryless nonlinearity. It then uses a uniform quantizer. The receiver applies the inverse expansion.

THE TWO STANDARD COMPRESSORS

$$y = \frac{\ln(1+\mu|x|)}{\ln(1+\mu)}\,\mathrm{sgn}(x), \qquad |x| \le 1$$

$$y = \begin{cases} \dfrac{A|x|}{1+\ln A}\,\mathrm{sgn}(x), & 0 \le |x| \le 1/A \\ \dfrac{1+\ln(A|x|)}{1+\ln A}\,\mathrm{sgn}(x), & 1/A < |x| \le 1 \end{cases}$$

Both laws map $x = \pm 1$ to $y = \pm 1$. The standard parameters are $\mu = 255$ and $A = 87.6$. For the same speech and eight-bit quantizer, their SQNR values differ by less than 0.01 dB.



The identity line shows the result without companding. Both compressors use much of the output range for small input amplitudes.

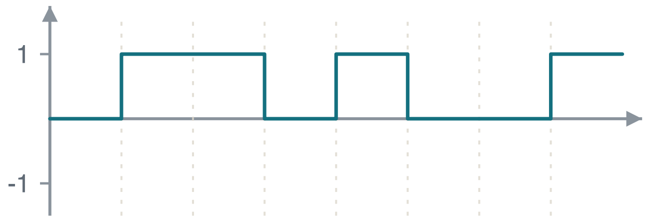
1.6 Encoding, line codes and pulse code modulation

With $L = 2^R$ levels, each sample needs R bits. At f_s samples per second, the bit rate is $R_b = Rf_s$.

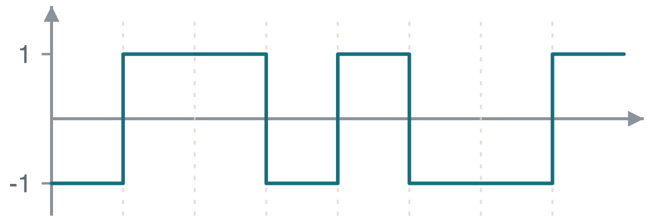
Natural binary coding assigns increasing binary values to the levels. Adjacent **Gray** words differ by one bit. Thus, a small level error changes only one Gray-coded bit.

A **line code** converts the bits into a waveform. Unipolar NRZ has a DC component that causes droop in an AC-coupled stage. Balanced polar NRZ has no DC component.

A long run of equal NRZ bits has no transitions for clock recovery. Manchester coding adds a transition to every bit but uses twice the bandwidth.



Unipolar NRZ.



Polar NRZ, carrying the same eight bits 01101001.

EXAMPLE 1.4 — A COMPLETE PCM STREAM

GIVEN $m(t) = 8|\operatorname{sinc}(t-2)|$, so $0 \leq m(t) \leq 8$, sampled every $T_s = 0.6$ s and quantized by an eight-level uniform quantizer covering $[0, 8]$.

FIND The step size, the levels, the code words for $t = 0, 0.6, \ldots, 3.6$, and the bit rate.

METHOD Calculate Δ from the range and level count. Evaluate each sample. Select the word from the quantization region.

SOLUTION $\Delta = 8/8 = 1$ V and the levels sit at $0.5, 1.5, \ldots, 7.5$. The samples are $0,\ 1.73,\ 1.87,\ 7.48,\ 6.05,\ 0,\ 1.51$, giving the words $000\ 001\ 001\ 111\ 110\ 000\ 001$. With $R = 3$ and $f_s = 1/0.6 = 1.667$, $R_b = 5$ bit/s and $T_b = T_s/3 = 0.2$ s.

CHECK At $t = 3$, the sample is $8|\operatorname{sinc}(1)| = 0$. The sinc function is zero at every nonzero integer. The interpolation formula uses the same property.

1.7 Vector quantization

Scalar quantization processes one sample at a time. It does not use the dependence between neighboring samples.

Vector quantization treats n samples as one point in n dimensions. A **codebook** contains the allowed output points. The quantizer selects the nearest codebook point.

EXAMPLE 1.5 — PAIRS OF NEIGHBOURING SAMPLES

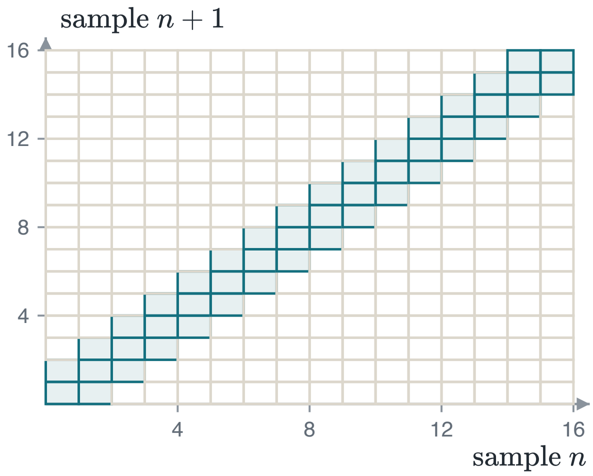
GIVEN — $L = 16$ levels, and a signal smooth enough that a sample never moves more than one step from the one before it.

FIND — What a pair costs, quantized separately and quantized together.

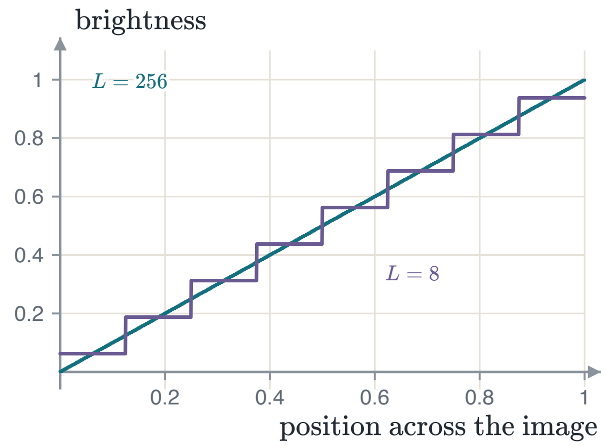
METHOD — Count the pairs each scheme has to be able to name, and take the base-two logarithm.

SOLUTION — Separately, every combination is allowed: $L^2 = 256$ pairs, $\log_2 256 = 8$ bits a pair, or 4 bits a sample. Together, only the pairs with $|i - j| \leq 1$ can occur, and there are $3L - 2 = 46$ of them: $\lceil \log_2 46 \rceil = 6$ bits a pair, or 3 bits a sample.

CHECK — The cell size and rounding error do not change. The code omits 210 pairs that the signal model cannot produce. The rate decreases from four to three bits per sample.



A scalar quantizer represents all 256 pairs. Under the smooth-signal assumption, only the 46 shaded pairs can occur.



The coarse quantizer changes a smooth gradient into flat steps. Each step boundary appears as a false line.

> **SOURCE DEPENDENCE**
>
> Most of the rate reduction comes from dependence between neighboring samples. Better cell shapes can give a smaller gain for independent samples. This course does not develop that case.

EXAMPLE 1.6 — QUANTIZING AN IMAGE

GIVEN — A 512×512 greyscale image at 8 bits a pixel, so $L = 256$ levels.

FIND — The size of the file, and the size and the cost at $L = 32$.

SOLUTION — $512^2(8) = 2\,097\,152$ bits, which is 256 KiB. At $L = 32$ the rate is $R = \log_2 32 = 5$ bits a pixel, so $1\,310\,720$ bits or 160 KiB — 37.5% smaller.

COST — $\Delta\text{SQNR} = 6.02(8 - 5) = 18.06$ dB. A smooth gradient becomes flat bands with false edges.

CHECK — The decoder cannot recover the position of a pixel inside its quantization interval. Therefore, this compression is **lossy**. JPEG also quantizes transformed image blocks.

1.8 Summary

RESULT	STATEMENT	ANCHOR
Replication	$G_\delta(f) = f_s \sum_n G(f - nf_s)$	PS CH7.1.1
Sampling theorem	$f_s \geq 2W$ for a message bandlimited to W	PS CH7.1.1
Reconstruction	$g_r(t) = \sum_n g(nT_s)\,\mathrm{sinc}(2Wt - n)$, filter gain $1/(2W)$	PS CH7.1.1
Step size	$\Delta = 2m_{\max}/L,\ \ L = 2^R$	PS CH7.2.1
Error power	$E[Q^2] = \Delta^2/12$, when the step is small	PS CH7.2.1
Signal-to-noise	$\mathrm{SQNR}\ [\mathrm{dB}] = \alpha + 6.02R$	PS CH7.2.1
Bit rate	$R_b = Rf_s$	PS CH7.3, 7.4.1
Vector quantization	round n samples together and use a codebook	PS CH7.2.2

Sampling is reversible and quantization is not. Everything after this chapter takes the bit stream as given and asks what the channel does to it.

CHAPTER 2

Baseband transmission of digital signals

Chapter 1 produced a stream of bits. Each bit now becomes a waveform, and the channel adds noise. The receiver must identify the transmitted waveform. This chapter selects the receiver filter and decision threshold. It also studies pulse overlap in a bandlimited channel.

2.1 The matched filter

Over one bit interval the receiver sees $x(t) = g(t) + w(t)$, the waveform plus white Gaussian noise of two-sided density $N_0/2$. It passes this through a filter and takes one sample at the end of the interval. Because the filter is linear, its output splits the same way, into a signal part $g_0(t) = g * h$ and a noise part $n(t) = w * h$.

The quantity worth maximising is the signal at the sampling instant against the noise power that accompanies it.

THE PEAK PULSE SIGNAL-TO-NOISE RATIO

$$(\mathrm{SNR})_o = \frac{|g_0(T)|^2}{E[n^2(t)]} = \frac{\left|\int_{-\infty}^{\infty} G(f)H(f)e^{j2\pi fT}df\right|^2}{\frac{N_0}{2}\int_{-\infty}^{\infty}|H(f)|^2 df}$$

Scaling H up multiplies the top and the bottom by the same factor, so the answer cannot be "make H large". What is being chosen is the shape of H.

Schwarz's inequality says that $\left|\int \phi_1\phi_2\right|^2 \leq \int |\phi_1|^2 \int |\phi_2|^2$, with equality only when $\phi_1 = k\phi_2^*$. Taking $\phi_1 = H(f)$ and $\phi_2 = G(f)e^{j2\pi fT}$, the factor $\int |H|^2$ cancels between the top and the bottom and a bound appears that contains no H at all.

THE BOUND, AND THE FILTER THAT REACHES IT

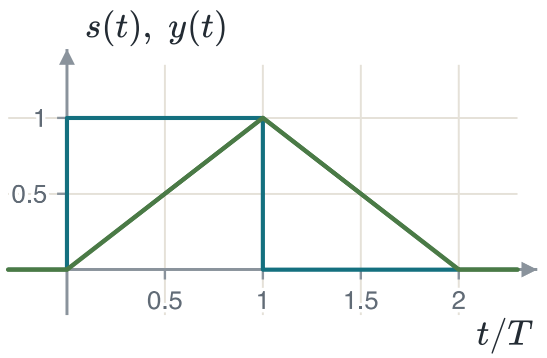
$$(\mathrm{SNR})_o \le \frac{2}{N_0}\int |G(f)|^2 df = \frac{2E}{N_0}$$

$$H(f) = k\,G^*(f)e^{-j2\pi fT} \quad \Longleftrightarrow \quad h_{\mathrm{opt}}(t) = k\,g(T-t)$$

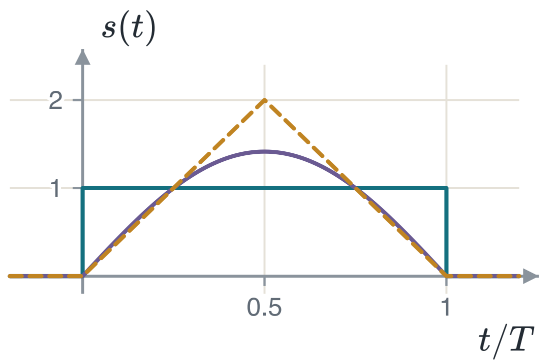
The optimum filter is the transmitted waveform reversed in time and shifted into the interval. It is called the **matched filter**.

TWO THINGS THIS SAYS

The best achievable ratio is $2E/N_0$. It depends on the **energy** of the pulse and on the noise density, and on nothing else about the pulse. Two completely different waveforms of the same energy perform identically. The constant k cancels. Therefore, the filter is fixed only up to a gain.



A rectangular pulse and the output of the filter matched to it. The output peaks at exactly $t = T$, and its peak is the energy of the pulse.



Three pulses of the same energy. Against white Gaussian noise a matched-filter receiver performs identically on all three.

2.2 From waveform to number

Both waveforms of a binary baseband system are multiples of one shape, so a single unit-energy function carries both. With $\psi(t) = 1/\sqrt{T_b}$ on the bit interval, polar NRZ is $s_m(t) = s_m\psi(t)$ with $s_0 = -A\sqrt{T_b}$ and $s_1 = +A\sqrt{T_b}$. Both carry energy $E_b = A^2T_b$, so $A\sqrt{T_b} = \sqrt{E_b}$ and the two waveforms have become two *numbers*, $\pm\sqrt{E_b}$, on one axis.

The demodulator can be built two ways. The **matched filter** convolves with $\psi(T_b - t)$ and samples at T_b. The **correlator** multiplies by $\psi(t)$ and integrates over the interval. At the sampling instant the two produce the same number. This occurs because convolving with a reversed function and evaluating at the end of the interval *is* correlating over it.

THE DECISION STATISTIC

$$y = \int_0^{T_b} x(\tau)\psi(\tau)\,d\tau = s_m\underbrace{\int_0^{T_b}\psi^2}_{=1} + \underbrace{\int_0^{T_b} w\psi}_{n} = s_m + n$$

They agree at $t = T_b$ and nowhere else, which is one more reason the sampling instant matters.

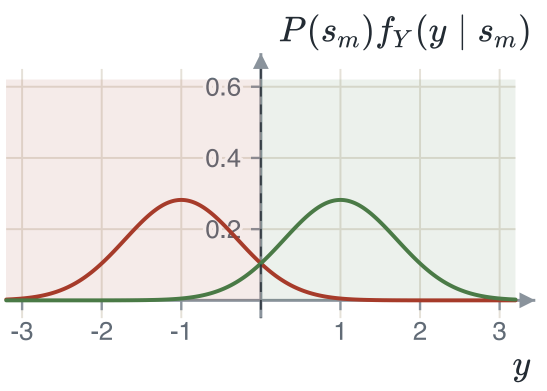
2.3 The decision and its error probability

The noise term is a projection of a Gaussian process onto a fixed function, so it is a Gaussian random variable. Its variance follows from $E[w(\tau)w(u)] = \frac{N_0}{2}\delta(\tau - u)$ and the sifting property, and the unit energy of ψ makes the answer independent of the interval length.

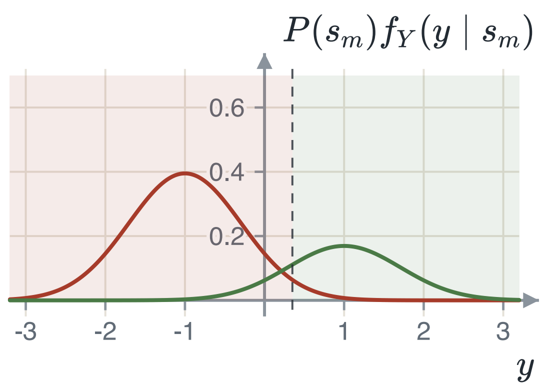
> WHAT THE DETECTOR IS GIVEN
>
> $$\sigma_n^2 = \frac{N_0}{2}\int_0^{T_b} \psi^2(\tau)d\tau = \frac{N_0}{2}$$
>
> $$y = s_m + n \sim \mathcal{N}\left(s_m, \frac{N_0}{2}\right)$$
>
> The two-sided convention arrives here unchanged. Reading the density as one-sided makes every error probability in the course 3 dB too optimistic, and nothing in the algebra shows it.

The detector decides "1" when $y > \lambda$. Write the average error as two integrals. Differentiate with respect to λ by the Leibniz rule. Set the derivative to zero. The result places the threshold where the two prior-weighted densities cross.

> THE OPTIMAL THRESHOLD
>
> $$P(s_1)f_Y(\lambda \mid s_1) = P(s_0)f_Y(\lambda \mid s_0)$$
>
> $$\lambda_{\text{opt}} = \frac{N_0}{4\sqrt{E_b}} \ln \frac{P(s_0)}{P(s_1)}$$
>
> Equal priors put the threshold at the midpoint. If s_0 is more likely, the threshold moves in the positive direction. This movement enlarges the decision region for s_0.



Equal priors: the threshold sits midway and the shaded decision regions are symmetric.



$P(s_0) = 0.7$: the more likely symbol's curve is taller and the crossing has moved towards the less likely one.

With equal priors the threshold is at the origin and both conditional errors are the same Gaussian tail, using $P(Y > y) = Q\left(\frac{y-\mu}{\sigma}\right)$ and $Q(-x) = 1 - Q(x)$.

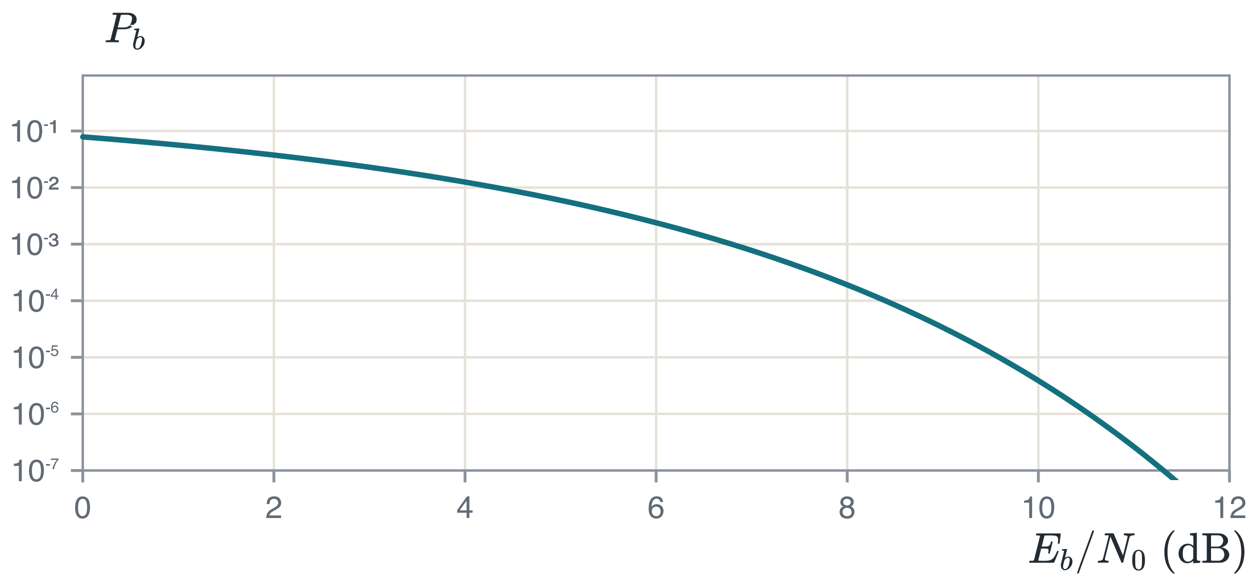
THE RESULT OF THE CHAPTER

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

It depends on E_b/N_0 and on nothing else — not on the amplitude, not on the bit duration, not on the pulse shape. Doubling the amplitude and quartering the duration change neither the energy per bit nor the answer.

ANTIPODAL AND ORTHOGONAL SIGNALS

Antipodal signaling gives $Q\left(\sqrt{2E_b/N_0}\right)$. On-off or orthogonal signaling gives $Q\left(\sqrt{E_b/N_0}\right)$ and needs 3 dB more energy. For on-off signaling, half the symbols have zero energy. Therefore, average bit energy is half the nonzero-symbol energy.



Bit error probability against E_b/N_0. Past about 8 dB every extra decibel is worth roughly an order of magnitude in error rate.

EXAMPLE 2.1 — UNEQUAL PRIORS

GIVEN A binary PAM system with correlator output $y = \pm\sqrt{E_b} + n$, where $P(s_1) = 0.3$, $E_b = 1$ and $N_0 = 0.1$.

FIND The optimal threshold and the average error probability there.

METHOD Threshold from the log-ratio of the priors. Each conditional error from a Gaussian tail. Average by weighting.

SOLUTION $\lambda = \frac{0.1}{4} \ln \frac{0.7}{0.3} = 0.0212$ and $\sigma = \sqrt{0.05} = 0.2236$. Then $P(\text{err} \mid s_0) = Q(4.567) = 2.475 \times 10^{-6}$ and $P(\text{err} \mid s_1) = Q(4.377) = 6.005 \times 10^{-6}$, so $P_e = 0.7(2.475) + 0.3(6.005)$ in units of 10^{-6}, giving 3.534×10^{-6}.

CHECK Leaving the threshold at zero would give $Q(\sqrt{20}) = 3.872 \times 10^{-6}$. Moving it improves the answer by about nine per cent. A real gain and a small one. This is what a shift of a tenth of a standard deviation buys.

2.4 Intersymbol interference

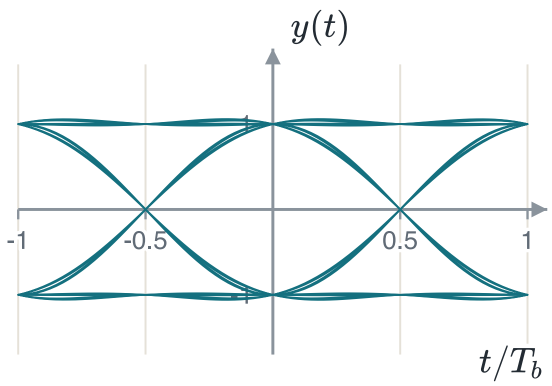
Everything above assumed an ideal channel. A real bandlimited channel spreads each pulse in time. Therefore, one pulse can enter adjacent symbol intervals. Write the received signal as a train of the overall pulse p. Sampling at $t_i = iT_b$ separates the wanted pulse from intersymbol interference.

WHERE THE INTERFERENCE COMES FROM

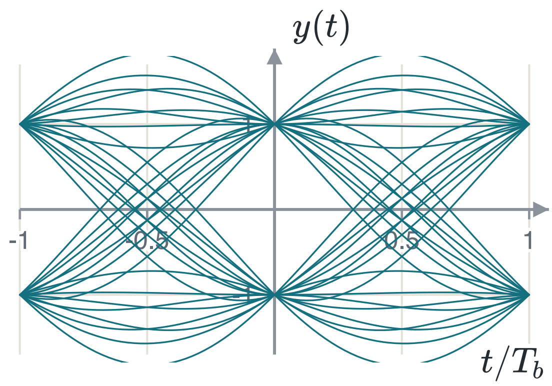
$$y(t_i) = \underbrace{\mu a_i}_{\text{wanted}} + \underbrace{\mu \sum_{k \neq i} a_k\, p\big((i-k)T_b\big)}_{\text{intersymbol interference}} + n(t_i)$$

The middle term is not noise. It is caused by the data itself, and raising the transmit power raises it by the same factor, so more power does not help. At high signal-to-noise ratio it is the only thing limiting the system.

An **eye pattern** shows this interference. Divide the received waveform into bit-length segments. Align the segments with the clock and draw them together. The opening height gives the noise margin. Its width gives the permitted sampling-time error. Crossing slope shows timing sensitivity, and crossing spread shows timing jitter. A closed eye has no sampling time that gives correct decisions for every bit pattern.



A wide-open eye: a pulse that decays quickly. Every trace passes close to ± 1 at the centre.



A slowly decaying pulse. The opening has narrowed and the crossings have scattered, with the noise unchanged.

2.5 Nyquist's criterion and the raised cosine

The interference vanishes exactly when the overall pulse is zero at every sampling instant but its own. Sampling p at those instants and transforming turns that requirement into a statement about the spectrum.

NYQUIST'S CRITERION FOR DISTORTIONLESS TRANSMISSION

$$p\big((i-k)T_b\big) = \begin{cases} 1, & i = k \\ 0, & i \neq k \end{cases}$$

$$R_b \sum_n P(f - nR_b) = 1, \qquad R_b = \frac{1}{T_b}$$

In words: the replicas of the pulse spectrum, spaced by the symbol rate, must add to a constant. The simplest spectrum that does it is a rectangle of width $2W$ with $W = R_b/2$, the **Nyquist bandwidth**, whose pulse is $\mathrm{sinc}(2Wt)$.

THE RATE A BANDWIDTH SUPPORTS

A channel of bandwidth W carries at most $2W$ symbols per second with no interference. This is the counterpart of the sampling theorem, reached from the other end of the same argument.

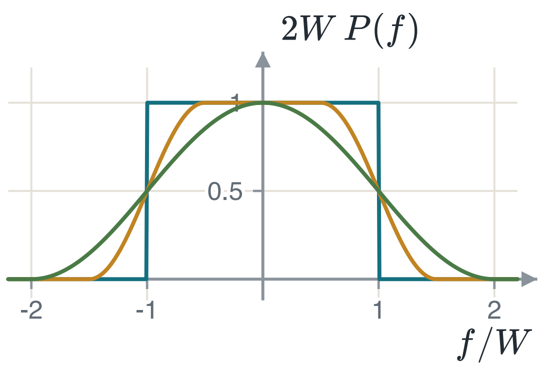
The ideal Nyquist channel cannot be built. Its spectrum has vertical edges, and its pulse decays only as $1/t$. Therefore, a small timing error lets a long tail of neighbours contribute at once. The fix is to widen the spectrum and taper its edges while keeping the tiling property.

THE RAISED COSINE

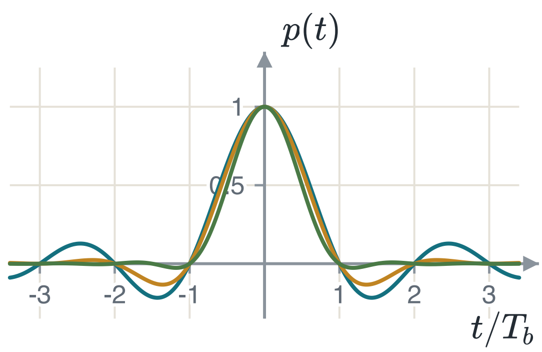
$$P(f) = \begin{cases} \dfrac{1}{2W}, & 0 \le |f| \le f_1 \\ \dfrac{1}{4W}\left[1 - \sin\dfrac{\pi(|f| - W)}{2W - 2f_1}\right], & f_1 \le |f| < 2W - f_1 \\ 0, & \text{otherwise} \end{cases}$$

$$p(t) = \mathrm{sinc}(2Wt)\,\frac{\cos(2\pi\alpha W t)}{1 - 16\alpha^2 W^2 t^2}, \qquad \alpha = 1 - \frac{f_1}{W}$$

The first factor keeps the zero crossings at $t = iT_b$, so the criterion still holds. The second decays as $1/t^2$. The price is bandwidth: $B_T = (1+\alpha)W$, an excess of αW over Nyquist.



The spectrum at $\alpha = 0$, 0.5 and 1. All three tile the axis at spacing $2W$, so all three give zero interference.



Their pulses. All three vanish at every non-zero multiple of T_b. The tails differ by orders of magnitude.

EXAMPLE 2.2 — FITTING A RATE INTO A CHANNEL

GIVEN	A channel of bandwidth 48 kHz is to carry 64 kbit/s with raised-cosine shaping.
FIND	The Nyquist bandwidth and the largest roll-off that fits.
METHOD	$W = R_b/2$, then $(1+\alpha)W \le B$.
SOLUTION	$W = 32$ kHz, and $(1+\alpha)(32) \le 48$ gives $\alpha \le 0.5$. At $\alpha = 0.5$ the signal uses the whole 48 kHz, with an excess of 16 kHz over the Nyquist bandwidth.
CHECK	At $\alpha = 0$, the signal needs only 32 kHz and still fits. However, its pulse decays as $1/t$, and its ideal filter cannot be built. Using the available bandwidth gives a pulse that decays as $1/t^3$. The spectral efficiency is $64/48 = 1.33$ bit/s/Hz, below the theoretical maximum of 2.

2.6 Summary

RESULT	STATEMENT	ANCHOR
Matched filter	$h_{\text{opt}}(t) = g(T-t)$	PS CH8.3.2
What it achieves	$(\text{SNR})_o = 2E/N_0$, independent of the pulse shape	PS CH8.3.2
Decision statistic	$y = s_m + n,\ n \sim \mathcal{N}(0, N_0/2)$	PS CH8.3.1
Optimal threshold	$\lambda = \frac{N_0}{4\sqrt{E_b}} \ln \frac{P(s_0)}{P(s_1)}$	PS CH8.3.3
Antipodal error	$P_b = Q\left(\sqrt{2E_b/N_0}\right)$	PS CH8.3.3
On-off error	$P_b = Q\left(\sqrt{E_b/N_0}\right)$, three decibels worse	PS CH8.3.3
Nyquist criterion	$R_b \sum_n P(f - nR_b) = 1$	PS CH10.3.1
Raised cosine	$B_T = (1+\alpha)W$	PS CH10.3.1

Two waveforms have become two points on a line, and the error probability has turned out to depend on the distance between them. Chapter 3 asks what happens when there are more than two waveforms, and finds that the same picture works with more axes.

CHAPTER 3

Geometric representation of signal waveforms

This chapter changes the language, not the subject. A set of waveforms is written as a set of points. Once that is done, the receiver becomes a question about geometry, and the answers are arithmetic instead of integrals.

3.1 Why one axis is not enough

In Chapter 2 the two waveforms were opposites. One shape carried both, the receiver produced one number, and everything followed from that. Now suppose the two waveforms are simply different: neither is a multiple of the other. No single function can write both as a multiple of itself, so the receiver of Chapter 2 does not apply.

The fix is to use two matched filters, one for each waveform, and let the receiver work with the two numbers they produce. A pair of numbers is a point in a plane. That plane is the subject of this chapter.

3.2 Signals as vectors

A vector is a list of numbers because it has been written against a set of axes. A signal can be a list of numbers for the same reason. The only thing needed is an inner product, and for signals it is the integral of the product.

INNER PRODUCT, ENERGY AND ORTHOGONALITY

$$\langle x, y\rangle = \int_{-\infty}^{\infty} x(t)\, y(t)\, dt$$

$$\int_{-\infty}^{\infty} \psi_j(t)\psi_k(t)\, dt = \begin{cases} 1, & j = k \\ 0, & j \neq k \end{cases}$$

The second line defines an **orthonormal** set. The first case says each function has unit energy. The second says any two of them are orthogonal.

ORTHOGONAL SIGNALS

Orthogonal signals have zero inner product. Multiply the signals and integrate the result over all time. Nonoverlapping pulses are a simple example. Where one pulse is nonzero, the other is zero. Therefore, their product is zero everywhere.

With the axes fixed, a signal is written against them the same way a vector is. One integral per axis takes the waveform apart. Adding the pieces back builds it again.

COORDINATES

$$s_{ij} = \int_0^T s_i(t)\, \psi_j(t)\, dt$$

$$s_i(t) = \sum_{j=1}^{N} s_{ij}\, \psi_j(t)$$

The list $\mathbf{s}_i = (s_{i1}, \ldots, s_{iN})$ is the **signal vector**.

The reason this is worth doing is that inner products survive the translation. Expand both signals in the basis, exchange the integral with the sums, and use orthonormality: every cross term vanishes and every matching term contributes one. An integral over waveforms becomes a sum over N numbers.

THE PROPERTY THE CHAPTER RESTS ON

$$\int_{-\infty}^{\infty} x(t)y(t)\,dt = \sum_{k=1}^{N} x_k y_k$$

$$E_{s_i} = \|\mathbf{s}_i\|^2 = \sum_{j=1}^{N} s_{ij}^2, \qquad \|\mathbf{s}_i - \mathbf{s}_k\|^2 = \int_0^T \left(s_i - s_k\right)^2 dt$$

Energy is how far a point is from the origin. That is what the transmitter pays for. **Distance is how far two points are from each other.** That is what decides how often the receiver confuses them.

3.3 The constellation diagram

The constellation is the picture of the signal vectors in the space their basis spans: one point per waveform, one axis per basis function. Three things are read straight off it. The distance from the origin to a point is the square root of that signal's energy. The distance between two points is the square root of the energy of their difference. The number of axes is the number of matched filters the receiver needs.

GEOMETRIC EQUIVALENCE

Different waveform sets can have the same constellation. Such sets have the same receiver structure and error probability in white Gaussian noise. The constellation does not determine bandwidth. Waveform shape and the Chapter 2 criteria determine bandwidth.

3.4 The Gram–Schmidt procedure

The previous sections assumed known axes. Gram–Schmidt produces them from any waveform set. Take the next signal. Remove its components along the existing axes. Normalize the nonzero remainder to form a new axis.

THE PROCEDURE

$$\psi_1(t) = \frac{s_1(t)}{\sqrt{E_1}}, \qquad s_{11} = \sqrt{E_1}$$

$$g_k(t) = s_k(t) - \sum_{i=1}^{k-1} s_{ki}\psi_i(t), \qquad \psi_k(t) = \frac{g_k(t)}{\sqrt{E_{g_k}}}$$

If $g_k(t) = 0$ the signal was already a combination of the axes found so far and **no new axis is added**. The number of axes N is therefore at most the number of signals M, and is often fewer.

THE ORDER CHANGES THE AXES, NOT THE ANSWER

Starting from a different signal gives a different basis. It gives the same number of axes, the same energies and the same distances. Therefore, the constellation is the same picture seen from a different angle. Every result that depends only on distances is unchanged. In Chapter 4 every result depends only on distances.

EXAMPLE 3.1 — THREE PULSES, TWO AXES

GIVEN $s_1(t) = 1$ on $[0,2]$; $s_2(t) = 1$ on $[2,3]$; $s_3(t) = 1$ on $[0,3]$; each zero elsewhere.

FIND An orthonormal basis, the three signal vectors, and the constellation.

METHOD Normalise the first signal. Then, for each later signal, subtract the part that lies along the axes already found and normalise the remainder.

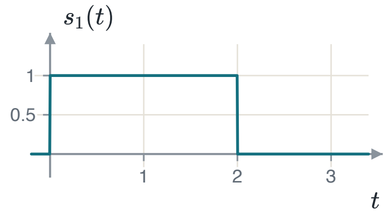
STEP 1 $E_1 = \int_0^2 1\,dt = 2$, so $\psi_1(t) = 1/\sqrt{2}$ on $[0,2]$ and $s_{11} = \sqrt{2}$.

STEP 2 $s_{21} = \int s_2\psi_1\,dt = 0$, because the two never overlap. So $g_2 = s_2$, its energy is 1, and $\psi_2(t) = s_2(t)$.

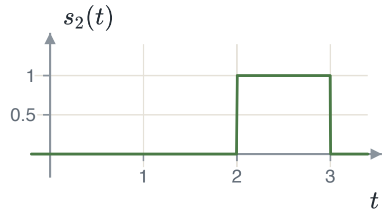
STEP 3 $s_{31} = \sqrt{2}$ and $s_{32} = 1$, and then $g_3 = s_3 - \sqrt{2}\psi_1 - \psi_2 = 0$. No third axis is added.

VECTORS $\mathbf{s}_1 = (\sqrt{2}, 0)$, $\mathbf{s}_2 = (0,1)$, $\mathbf{s}_3 = (\sqrt{2}, 1)$.

CHECK Energies from the vectors are 2, 1 and 3. From the waveforms: height one for two seconds, one second and three seconds, giving 2, 1 and 3. They agree, and that agreement is the whole content of "energy is squared length". Three waveforms needed only two axes, because $s_3 = s_1 + s_2$ — visible in the pictures before any integral is computed.

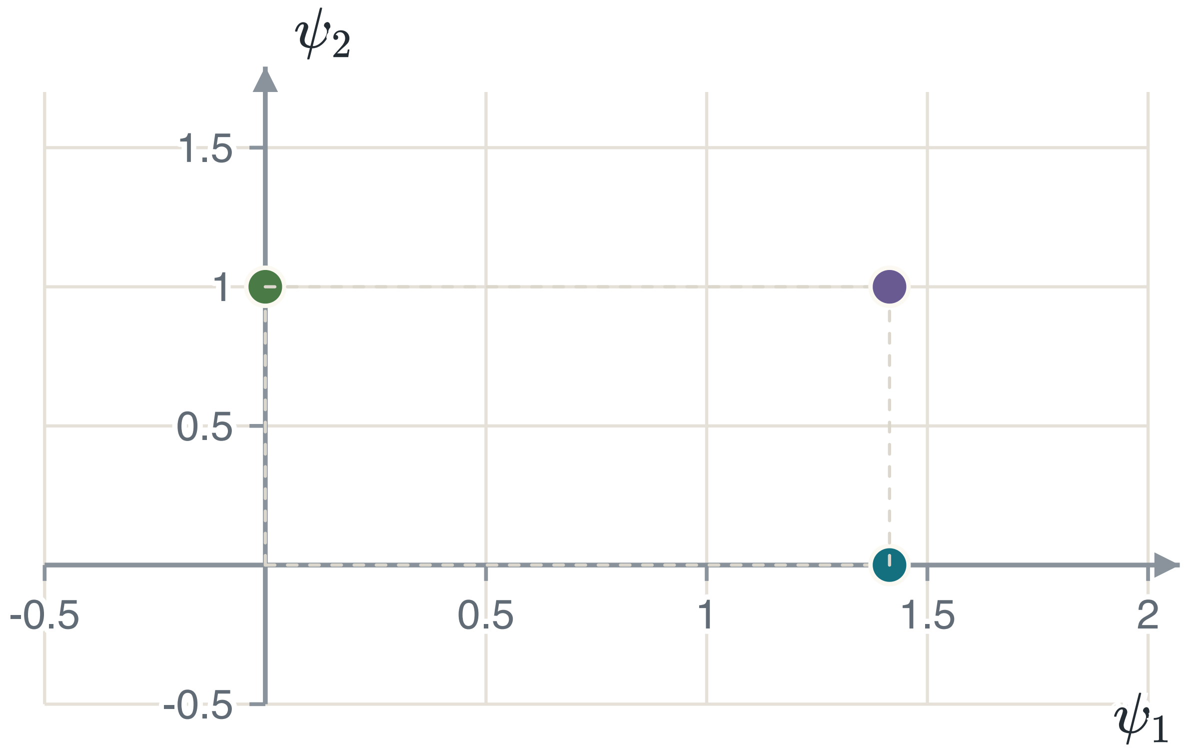


s_1



s_2

$s_3 = s_1 + s_2$



The constellation. Two axes carry three signals. The point furthest from the origin is $\mathbf{s}_3$, whose energy is 3. The two nearest points are $\mathbf{s}_1$ and $\mathbf{s}_3$, one unit apart.

3.5 Summary

RESULT	STATEMENT	ANCHOR
Orthonormal set	$\int \psi_j \psi_k \, dt = 1$ if $j = k$, else 0	PS CH8.1
Coordinates	$s_{ij} = \int s_i \psi_j \, dt$	PS CH8.1
Inner product survives	$\int xy \, dt = \sum_k x_k y_k$	PS CH8.1
Energy	$E_i = \|\mathbf{s}_i\|^2$	PS CH8.1
Distance	$\|\mathbf{s}_i - \mathbf{s}_k\|^2 =$ energy of the difference	PS CH8.1
Gram–Schmidt	$g_k = s_k - \sum_{i<k} s_{ki}\psi_i$, and $\psi_k = g_k / \sqrt{E_{g_k}}$	PS CH8.1

Chapter 4 builds the receiver for this constellation. It calculates the received coordinates and selects the nearest point. The error probability then depends on the distances between signal points.

CHAPTER 4

The optimal receiver in additive white Gaussian noise

A transmitter picks one of M signals and sends it. The channel adds noise. The receiver has to name the signal. Chapter 3 turned the signals into points. This chapter turns the receiver into a rule about those points, and the rule is simple: choose the point nearest to what arrived.

4.1 What the receiver keeps

The received signal is $r(t) = s_i(t) + n(t)$ over one symbol interval. The receiver has N correlators, one for each basis function, and each returns one number.

THE OBSERVATION VECTOR

$$r_k = \int_0^T r(t)\psi_k(t)\,dt = s_{ik} + n_k$$

$$\mathbf{r} = \mathbf{s}_i + \mathbf{n} = (r_1, \ldots, r_N)$$

The signal was built from the N basis functions, so the correlators capture all of it. The noise was not: the part of it outside the signal space, written $n_0(t)$, is thrown away.

NOISE OUTSIDE THE SIGNAL SPACE

The component $n_0(t)$ contains no signal because every signal lies in the basis span. It is also independent of the retained coordinates. Therefore, it cannot help the receiver identify the transmitted signal. The receiver still keeps all noise components inside the signal space.

Each n_k is a projection of a Gaussian process onto a fixed function, so it is Gaussian with zero mean. The useful part is what happens between two of them: using $E[n(\tau)n(u)] = \frac{N_0}{2}\delta(\tau - u)$ and orthonormality, $E[n_j n_k]$ is $N_0/2$ when $j = k$ and zero otherwise. Uncorrelated jointly Gaussian variables are independent, so the noise components are independent, each $\mathcal{N}(0, N_0/2)$.

NOISE GEOMETRY

Every axis has the same variance, and the axes are independent. Therefore, the noise cloud is circular in two dimensions and spherical in N dimensions. This symmetry gives the nearest-point decision rule.

4.2 The decision rule

The receiver should choose the signal that is most likely given what it saw. By Bayes' rule that means maximising $P(\mathbf{s}_i) f_{\mathbf{r}}(\mathbf{r} \mid \mathbf{s}_i)$ — the prior times the likelihood. This is the **MAP** rule, and no rule has a smaller error probability. If all M signals are equally likely the priors cannot change the answer, and what is left is the **ML** rule: maximise the likelihood alone.

Substitute the noise density into the likelihood. Then take its logarithm. The logarithm is increasing, so it does not change the maximizing signal. It converts the product into a sum and isolates the term that depends on i.

MINIMUM-DISTANCE DETECTION

$$\ln f_{\mathbf{r}}(\mathbf{r} \mid \mathbf{s}_i) = -\frac{N}{2}\ln(\pi N_0) - \frac{1}{N_0}\|\mathbf{r} - \mathbf{s}_i\|^2$$

$$\hat{s} = \arg\min_i \|\mathbf{r} - \mathbf{s}_i\|^2 \quad \text{(ML)}$$

$$\hat{s} = \arg\min_i \left\{ \|\mathbf{r} - \mathbf{s}_i\|^2 - N_0 \ln P(\mathbf{s}_i) \right\} \quad \text{(MAP)}$$

Select the signal point with the smallest metric. For unequal priors, the prior term changes each distance metric. A more likely symbol receives a larger decision region.

Expanding the squared distance gives the form a receiver is actually built in. The term $\|\mathbf{r}\|^2$ is the same for every i and drops out. Turning the minimum into a maximum and dividing by two leaves the **correlation metric**.

THE RECEIVER AS IT IS BUILT

$$\hat{s} = \arg\max_i \left\{ \mathbf{r} \cdot \mathbf{s}_i - \frac{E_i}{2} + \frac{N_0}{2}\ln P(\mathbf{s}_i) \right\}$$

And $\mathbf{r} \cdot \mathbf{s}_i = \int_0^T r(t) s_i(t)\, dt$, so the receiver can correlate against the waveforms directly and never compute coordinates. If the signals are equally likely **and** have equal energy, both corrections vanish and the rule is: take the largest correlation.

UNEQUAL SIGNAL ENERGIES

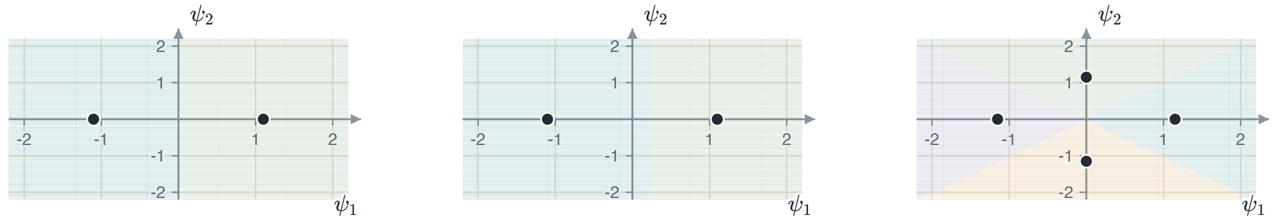
Keep $-E_i/2$ when signal energies differ. Without it, a large $\mathbf{s}_i$ makes the correlation metric favor a high-energy signal. This error makes an on-off receiver select "one" too often. Equal priors remove only the prior term.

4.3 Decision regions

Collecting the observations that give the same answer divides the space into M regions, $R_i = \{\mathbf{r} : \|\mathbf{r} - \mathbf{s}_i\| \leq \|\mathbf{r} - \mathbf{s}_j\| \text{ for all } j\}$. Three facts describe them, and all three follow from the rule being "nearest point".

1. A boundary between two points is **perpendicular** to the line joining them.
2. With equal priors it crosses that line exactly **halfway**.
3. With unequal priors it moves, and **the region of the less likely signal shrinks**.

A decision boundary is the perpendicular bisector of two signal points. Only nearest neighbors contribute boundaries. An interior point has a bounded decision region. An outer point has an unbounded region and usually fewer nearest neighbors.



Two points, equal priors: the boundary is halfway.

The left symbol four times more likely: its region has grown.

Four points, four boundaries, each perpendicular to the pair it separates.

For two equally likely points a distance d apart, the boundary is $d/2$ from each. An error happens when the noise along the line joining them carries the observation across, and the noise on any one axis is $\mathcal{N}(0, N_0/2)$.

THE BINARY RESULT, FROM THE PICTURE

$$P_e = Q\left(\frac{d/2}{\sqrt{N_0/2}}\right) = Q\left(\sqrt{\frac{d^2}{2N_0}}\right)$$

$$\mu = \frac{d}{2} + \frac{N_0}{2d} \ln \frac{P(\mathbf{s}_1)}{P(\mathbf{s}_0)}$$

The second line is where the boundary sits when the priors are unequal, measured from the first point along the line. Equal priors give $\mu = d/2$.

BINARY ERROR PROBABILITY

The binary error probability depends only on the distance between the two signal points. Antipodal points are $2\sqrt{E_b}$ apart. On-off or orthogonal points are $\sqrt{2E_b}$ apart. The factor $\sqrt{2}$ gives the 3 dB difference from Chapter 2.

4.4 The union bound

For more than two signals, the exact error is an N-dimensional integral over each decision region. A region can have many faces. A simple closed form is usually unavailable. The union bound gives a computable upper bound.

Suppose $\mathbf{s}_k$ was sent, and let A_{kj} be the event that the observation is closer to $\mathbf{s}_j$ than to $\mathbf{s}_k$. An error happens exactly when at least one of those occurs, and the probability of a union is at most the sum of the probabilities. Each term is a two-point question whose answer is already known.

THE UNION BOUND AND ITS THREE USABLE FORMS

$$P(\mathbf{s}_k \to \mathbf{s}_j) = Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right)$$

$$P_e \le \frac{1}{M}\sum_k \sum_{j\ne k} Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right)$$

$$P_e \le (M-1)Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right), \qquad P_e \approx N_{\min} Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right)$$

$N_{\min}$ is the *average* number of points at the minimum distance, and need not be an integer. The last form is the one used in practice.

One more tightening is available, and it costs nothing. To land nearer a different point the observation must leave its own region, and it leaves through a **face**. A piece of boundary shared with one particular neighbour. A point that shares no face with the region cannot be the first one reached. Therefore, the union need only run over the neighbours that bound the region. Write $\mathcal{N}(k)$ for those.

THE INTELLIGENT UNION BOUND

$$P(\text{error} \mid \mathbf{s}_k) \le \sum_{j\in\mathcal{N}(k)} Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right)$$

Still an upper bound, because leaving the region means crossing one of its faces. It is not the nearest-neighbour form: this one counts *faces* and bounds, that one counts *points at* $d_{\min}$ and approximates. They agree whenever every face sits at $d_{\min}$, and part company when a face is shared with a point further away.

UNION-BOUND GAP

Pairwise error events can overlap, so their sum can count one observation more than once. The bound can exceed one at low signal-to-noise ratio. At high ratios, the overlaps become small and the bound approaches the exact error probability.

EXAMPLE 4.1 — FOUR BOUNDS ON ONE CONSTELLATION

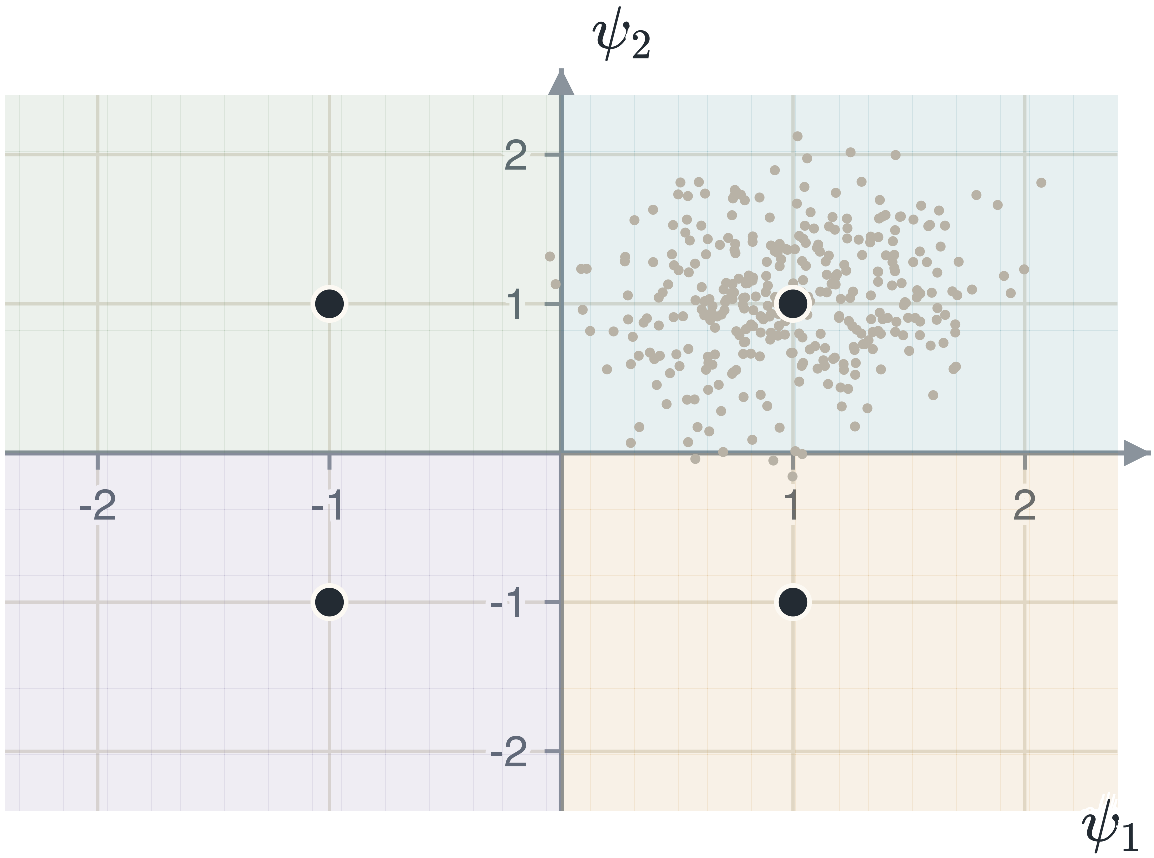
GIVEN Four equally likely points at $(\pm d/2, \pm d/2)$: neighbours d apart, diagonals $d\sqrt{2}$.

FIND The symbol error probability by each of the four forms.

METHOD Take one point, list its distances to the other three, and add one Q for each. Then ask which of those three actually bound its region. Symmetry makes every point give the same answer.

SOLUTION General: $P_e \le 2Q\left(\sqrt{d^2/2N_0}\right) + Q\left(\sqrt{2d^2/2N_0}\right)$. Intelligent (the region has two faces, both at d): $P_e \le 2Q\left(\sqrt{d^2/2N_0}\right)$. Nearest neighbours ($d_{\min} = d$, $N_{\min} = 2$): $P_e \approx 2Q\left(\sqrt{d^2/2N_0}\right)$. Minimum distance ($M - 1 = 3$): $P_e \le 3Q\left(\sqrt{d^2/2N_0}\right)$.

CHECK Put $d^2/2N_0 = 9$, so $Q(3) = 1.35 \times 10^{-3}$ and $Q(4.243) = 1.10 \times 10^{-5}$. The four answers are 2.71×10^{-3}, 2.70×10^{-3}, 2.70×10^{-3} and 4.05×10^{-3}: the diagonal term is worth 0.4%, and pretending it is a nearest neighbour costs 50%. The middle two agree here because both faces sit at $d_{\min}$, and only one of them is a bound. All four are the same Q with a different count in front, so none of them moves the curve sideways.



The constellation, its four regions, and the observations the receiver sees when the top-right point is sent. The exact error probability is the fraction of that cloud outside its own region. The bound adds three separate two-point questions and counts the overlaps twice.

4.5 Summary

RESULT	STATEMENT	ANCHOR
Observation	$\mathbf{r} = \mathbf{s}_i + \mathbf{n}$, each $n_k \sim \mathcal{N}(0, N_0/2)$, independent	PS CH8.4.1
MAP rule	minimise $\|\mathbf{r} - \mathbf{s}_i\|^2 - N_0 \ln P(\mathbf{s}_i)$	PS CH8.4.1
ML rule	minimise $\|\mathbf{r} - \mathbf{s}_i\|^2$: choose the nearest point	PS CH8.4.1
Receiver	maximise $\mathbf{r} \cdot \mathbf{s}_i - E_i/2 + \frac{N_0}{2} \ln P(\mathbf{s}_i)$	PS CH8.4.1
Regions	perpendicular bisectors. The less likely region shrinks	PS CH8.4.1
Binary	$P_e = Q\left(\sqrt{d^2/2N_0}\right)$	PS CH8.3.3
Union bound	$P_e \leq \frac{1}{M} \sum_k \sum_{j \neq k} Q(\cdot)$	PS CH8.4.2
Intelligent form	sum over the neighbours that give R_k a face	PS CH8.4.2
In practice	$P_e \approx N_{\min} Q\left(\sqrt{d_{\min}^2/2N_0}\right)$	PS CH8.4.2

Two parameters describe the nearest-neighbor approximation. The minimum distance $d_{\min}$ controls the Q -function argument. The average neighbor count $N_{\min}$ is a multiplier. Chapter 5 compares these values for practical constellations at fixed energy and bits per symbol.

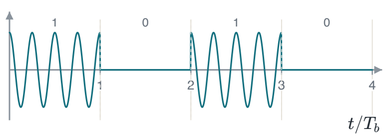
CHAPTER 5

Digital modulation methods

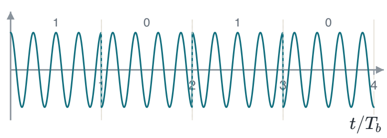
A baseband pulse cannot be sent over a radio channel, so the bits have to ride on a carrier. There are only three things about a sine wave that can be changed: its amplitude, its phase and its frequency. Every scheme in this chapter changes one of them, or two at once. Chapter 4 already gave the receiver and the error probability. All that is left is to place the points and measure the distance between them.

5.1 The three binary schemes

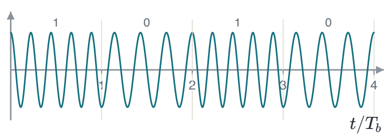
Take one bit at a time and two waveforms. Amplitude-shift keying sends the carrier for a one and nothing for a zero. Phase-shift keying sends the carrier either way and flips its sign. Frequency-shift keying sends one of two frequencies.



Amplitude-shift keying: the carrier is switched on and off.



Phase-shift keying: the carrier is always on and its sign flips.



Frequency-shift keying: one frequency for a one, another for a zero.

To compare them, put each on the axes of chapter 3 and measure. BPSK needs one basis function and its two points are $\pm\sqrt{E_b}$ on it. BFSK needs two, because the two frequencies are orthogonal, and its points sit one on each axis. BASK needs one, and its points are 0 and $\sqrt{2E_b}$ once the average over equally likely bits is taken.

THE THREE DISTANCES AND THE THREE ERROR PROBABILITIES

$$\text{BPSK: } d_{\min}^2 = 4E_b, \qquad P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

$$\text{BFSK and BASK: } d_{\min}^2 = 2E_b, \qquad P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$$

The two answers differ by a factor of two inside the square root. This is 3.01 dB. That gap does not depend on the noise level and cannot be closed by spending more power. It is a fact about where the points are.

THREE-DECIBEL DIFFERENCE

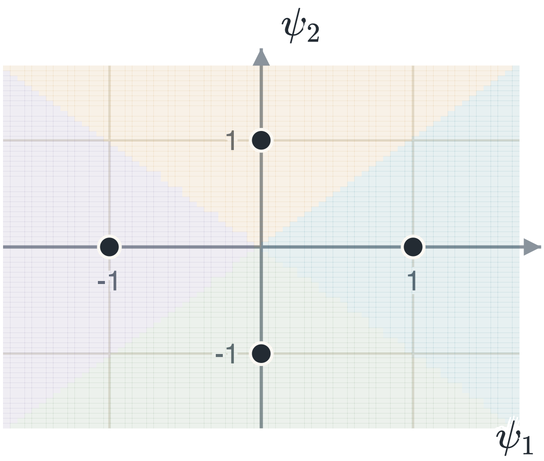
Antipodal points are $2\sqrt{E_b}$ apart. Orthogonal points are $\sqrt{2E_b}$ apart, which is smaller by $\sqrt{2}$. Squaring this ratio gives a factor of two. Therefore, the required energy differs by $10\log_{10} 2 = 3.01$ dB.

BASK AND BFSK REACH THE SAME ANSWER BY DIFFERENT ROUTES

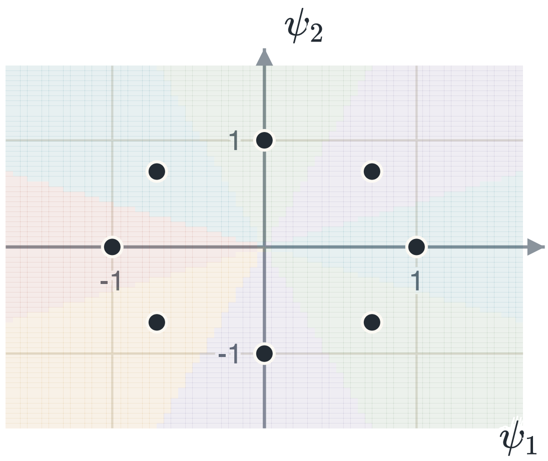
BASK keeps its points on one line but half its symbols cost nothing, so the average energy is half the peak. BFSK keeps the full energy in every symbol but separates the points by a right angle instead of a straight line. Both end at $d^2 = 2E_b$. The mathematics does not prefer either. The transmitter does, and BASK asks it to switch between zero power and full power.

5.2 M-ary phase-shift keying

Sending one bit at a time wastes the two dimensions a carrier offers. Use both, keep the energy fixed, and the constellation becomes M points spaced evenly around a circle of radius $\sqrt{E_s}$. Each symbol now carries $\log_2 M$ bits.



$M = 4$, two bits a symbol. The four regions are the four quadrants.



$M = 8$, three bits. The regions are wedges, and they are narrowing.

Two neighbouring points subtend an angle $2\pi/M$ at the centre. Dropping a perpendicular from the centre onto the line joining them cuts that angle in half and gives the distance directly.

M-PSK

$$d_{\min} = 2\sqrt{E_s}\,\sin\left(\frac{\pi}{M}\right), \qquad N_{\min} = 2$$

$$P_e \approx 2\,Q\left(\sqrt{\frac{2E_s}{N_0}}\,\sin\frac{\pi}{M}\right)$$

$N_{\min} = 2$ for every M above two, because a point on a circle has one neighbour clockwise and one anticlockwise. At $M = 2$ the formula gives $d_{\min} = 2\sqrt{E_s}$, which is BPSK — a general result that reproduces the case already known.

EXAMPLE 5.1 — WHAT THE STEP FROM FOUR POINTS TO EIGHT COSTS

GIVEN QPSK and 8-PSK, at whatever energy each needs to reach the same symbol error probability.

FIND The extra energy 8-PSK needs, per symbol and per bit.

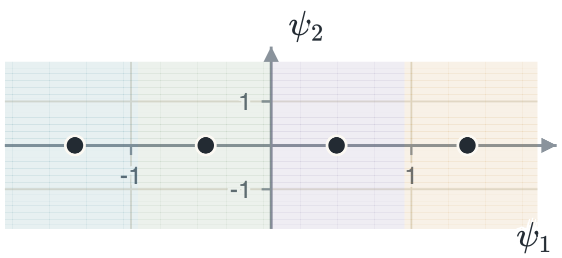
METHOD A fixed error probability means a fixed Q argument, so $(E_s/N_0)\sin^2(\pi/M)$ must be held constant. Take the ratio.

SOLUTION $\dfrac{\sin^2(\pi/4)}{\sin^2(\pi/8)} = \dfrac{0.5}{0.1464} = 3.414$, which is 5.33 dB per symbol. 8-PSK carries three bits where QPSK carries two, so per bit the figure is $3.414 \times \frac{2}{3} = 2.276$, or 3.57 dB.

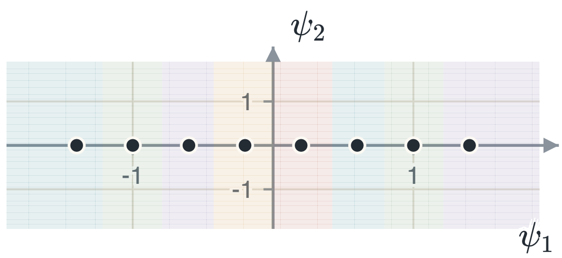
CHECK Both numbers are positive, as they must be: at the same radius, eight points are closer together than four. The per-bit figure is the smaller of the two, because the extra bit pays part of its own way.

5.3 M-ary amplitude-shift keying

Change the amplitude among M levels instead. One basis function carries them all, so the constellation is M points on a line at $\pm A, \pm 3A, \ldots, \pm(M-1)A$, and neighbouring points are $2A$ apart.



$M = 4$ on one axis. The two end regions run off to infinity.



$M = 8$: the same line, twice as crowded.

M-PAM

$$E_{s,\text{avg}} = \frac{A^2(M^2-1)}{3} \quad \Longrightarrow \quad d_{\min}^2 = \frac{12E_{s,\text{avg}}}{M^2-1}$$

$$N_{\min} = \frac{2(M-1)}{M}, \qquad P_e \approx N_{\min}\, Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right)$$

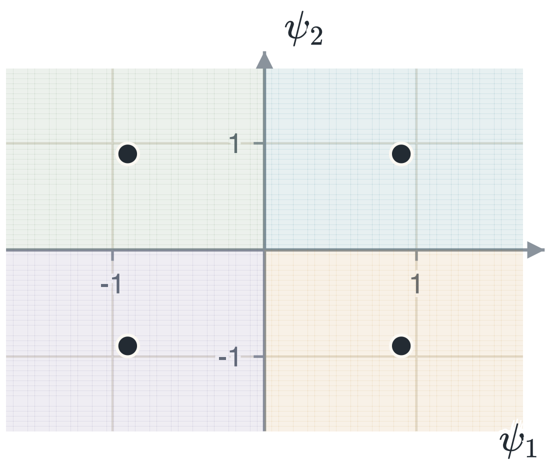
The M^2 in the denominator is the trouble. Doubling M divides the squared distance by roughly four. This is about 6 dB, and it does so at every doubling.

AVERAGE NEIGHBOR COUNT

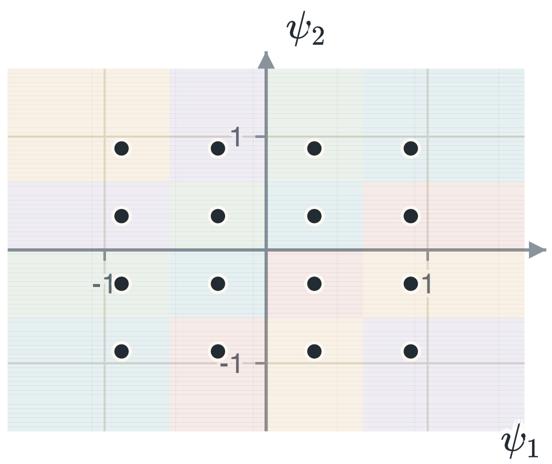
The two end points have one nearest neighbor each. The other $M-2$ points have two. Averaging over all equally likely symbols gives $N_{\min} = 2(M-1)/M$. Thus, $N_{\min} = 1.5$ for $M = 4$ and 1.75 for $M = 8$.

5.4 Quadrature amplitude modulation

Run one amplitude-modulated signal on the cosine and an independent one on the sine. The result is two $\sqrt{M}$-level constellations at right angles. Therefore, M points on a square grid. The lecture material states this as M-QAM being two-dimensional M-ASK.



4-QAM: two levels a side. These are the four points of QPSK — at $M = 4$ the two schemes are the same scheme.



16-QAM: four levels a side, four bits a symbol. The corners have two neighbours, the edges three, the four in the middle four.

SQUARE M-QAM

$$E_{s,\text{avg}} = \frac{(M-1)d^2}{6} \quad\Longrightarrow\quad d^2_{\min} = \frac{6E_{s,\text{avg}}}{M-1}$$

$$N_{\min} = \frac{4(2)+8(3)+4(4)}{16} = 3 \quad \text{for 16-QAM}$$

Compare the denominators: $M-1$ for QAM and M^2-1 for PAM. Two dimensions reduce the rate at which minimum distance decreases as M increases.

EXAMPLE 5.2 — SIXTEEN POINTS ON A GRID AGAINST SIXTEEN ON A LINE

GIVEN 16-QAM and 16-PAM at the same average symbol energy. Both carry four bits a symbol.

FIND The advantage of QAM in decibels.

METHOD One formula each, then the ratio of the squared distances.

SOLUTION QAM: $d^2_{\min} = 6E_s/15 = 0.400E_s$. PAM: $d^2_{\min} = 12E_s/255 = 0.0471E_s$. The ratio is 8.5, and $10\log_{10} 8.5 = 9.29$ dB.

CHECK Both methods carry the same number of bits at the same average power. Only their point geometry differs. The second dimension gives QAM a 9.3 dB distance advantage in this example.

5.5 M-ary frequency-shift keying

Use M frequencies, chosen so that the waveforms are orthogonal. Orthogonal signals need one basis function each. Therefore, the constellation lives in M dimensions with one point on each axis at distance $\sqrt{E_s}$ from the origin.

M-FSK

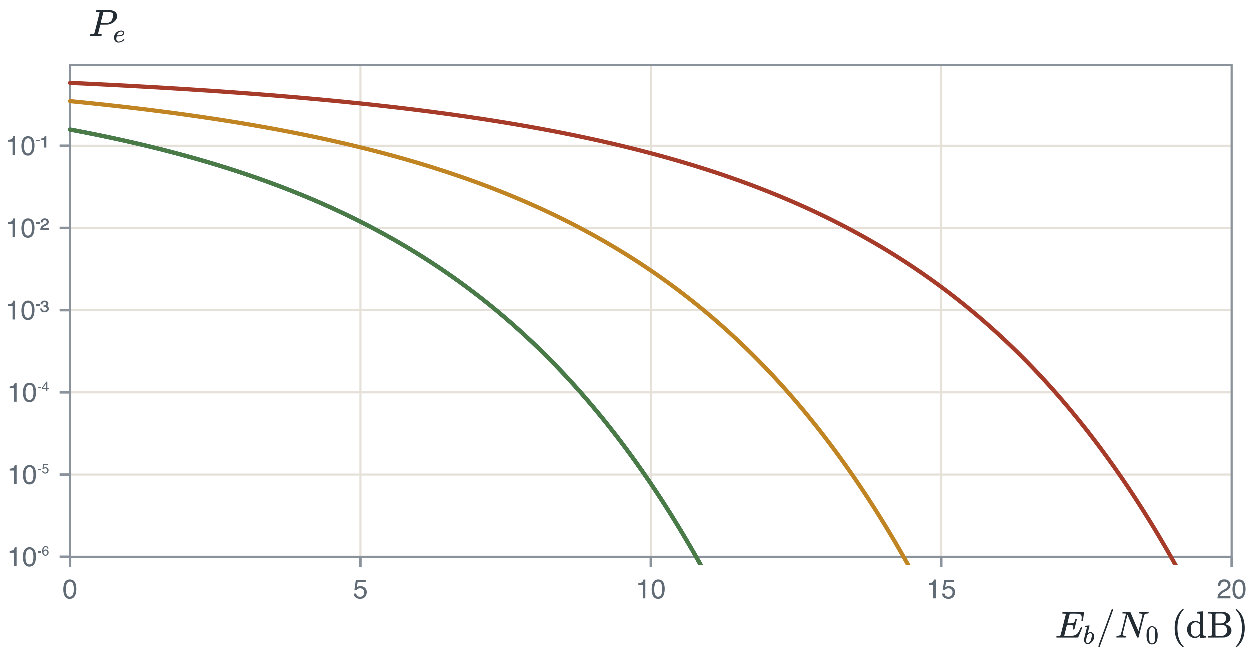
$$d_{jk} = \sqrt{2E_s} \;\; \text{for every pair,} \qquad N_{\min} = M-1$$

$$P_e \approx (M-1)\,Q\left(\sqrt{\frac{E_s}{N_0}}\right)$$

The distance does not depend on M at all. Adding waveforms costs nothing in distance — but each new waveform needs its own frequency slot, so the price is paid in bandwidth instead.

THE TRADE, STATED PLAINLY

PSK and QAM hold the bandwidth fixed as M grows and pay in energy, because the points crowd together. FSK holds the distance fixed and pays in bandwidth. A deep-space link, with bandwidth to spare and no power, chooses FSK. A mobile phone, with the opposite problem, chooses QAM. Neither is better in the abstract — it depends entirely on which resource is scarce.



Symbol error probability against energy per bit for four sizes of PSK. $M = 2, 4, 8, 16$ from left to right. $M = 2$ and $M = 4$ lie almost on top of each other. This is why QPSK is everywhere. It carries twice the bits of BPSK for the same energy per bit.

5.6 Summary

SCHEME	$d^2_{\min}$	$N_{\min}$	ANCHOR
BPSK	$4E_b$	1	PS CH8.6.1
BFSK, BASK	$2E_b$	1	PS CH9.5
M-PSK	$4E_s \sin^2(\pi/M)$	2	PS CH8.6.3
M-PAM	$12E_s/(M^2-1)$	$2(M-1)/M$	PS CH8.5.3
M-QAM	$6E_s/(M-1)$	3 at $M = 16$	PS CH8.7.1
M-FSK	$2E_s$	$M-1$	PS CH9.5

Every row was obtained the same way. Write the waveforms down, find the basis functions they need, place the points, measure the shortest distance, count how many pairs sit at it. The error probability then comes from chapter 4 without any further thought about waveforms. Convert once with $E_s = (\log_2 M)E_b$ when a question is stated per bit, and never convert twice.

What none of this says is how many bits a channel can carry at all. Every scheme here trades energy against bandwidth, and it is reasonable to ask whether there is a limit to the trade. Chapter 6 answers that, and the answer does not depend on which modulation is used.

CHAPTER 6

An introduction to information theory

The previous chapters started with a given bit stream and transmitted it through a channel. This chapter studies the source of those bits. Entropy measures the average information from the source. It also gives the minimum average rate for lossless source coding.

6.1 The information in one symbol

A **discrete memoryless source** emits one of K symbols each time, with fixed probabilities $p_1, \ldots, p_K$, chosen independently of what came before. How much is learnt when one arrives? If it was certain, nothing. If it was almost impossible, a great deal.

SELF-INFORMATION

$$I(s_k) = \log_a \frac{1}{p_k} = -\log_a p_k$$

The base sets the unit: $a = 2$ gives **bits**, $a = e$ gives **nats**, $a = 10$ gives **Hartleys**. This course uses base two throughout.

THREE PROPERTIES, AND WHAT FORCES THE LOGARITHM

$I(s_k) \geq 0$, because a probability is at most one and its logarithm at most zero.
$I(s_k) \geq I(s_j)$ **when** $p_k \leq p_j$: rarer means more informative.
$I(s_k s_j) = I(s_k) + I(s_j)$ **for independent symbols**, because their probabilities multiply and the logarithm turns a product into a sum. That third property is the one that rules out every function except a logarithm.

ONE HALVING IS ONE BIT

$-\log_2 \frac{1}{2} = 1$, so every halving of a probability adds exactly one bit. A symbol of probability $1/1000$ is about ten halvings away from certainty, so it carries about 10 bits. Most estimates in this chapter can be done that way, without a calculator.

6.2 Entropy

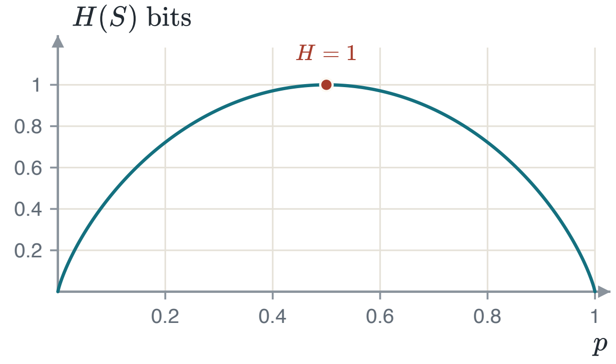
Self-information describes one symbol. Average it over the alphabet, weighting each symbol by how often it happens, and the result describes the source.

ENTROPY

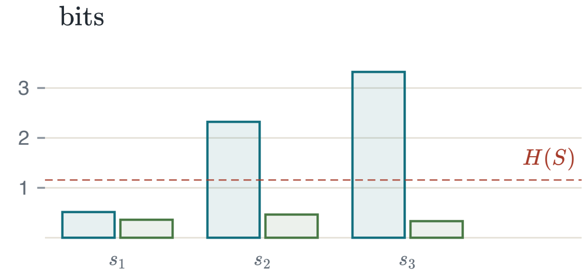
$$H(S) = E\big[I(s_k)\big] = -\sum_{k=1}^{K} p_k \log_2 p_k \quad \text{bits a symbol}$$

$$0 \leq H(S) \leq \log_2 K$$

Zero when one symbol has probability one — nothing is ever in doubt. At the top, $\log_2 K$, when all K are equally likely: putting $p_k = 1/K$ into the sum gives $\sum \frac{1}{K} \log_2 K = \log_2 K$.



The entropy of a binary source. One bit at $p = \frac{1}{2}$, nothing at either end, and flat near the top — a slightly unfair coin is almost as informative as a fair one.



The standard example. The left bar of each pair is what the symbol carries. The right bar is that weighted by how often it happens. The three right bars add to $H(S)$.

EXAMPLE 6.1 — THE ENTROPY OF A THREE-SYMBOL SOURCE

GIVEN $S = \{s_1, s_2, s_3\}$ with probabilities $0.7,\ 0.2,\ 0.1$.

FIND The entropy.

SOLUTION $H(S) = -0.7 \log_2 0.7 - 0.2 \log_2 0.2 - 0.1 \log_2 0.1 = 0.3602 + 0.4644 + 0.3322 = 1.1568$ bits a symbol.

CHECK $\log_2 3 = 1.585$, and 1.1568 is below it, as it must be for a source that is not uniform. Numbering the three symbols would cost 2 bits each, so plain numbering wastes almost 0.85 bits every symbol. Recovering that is the whole of source coding.

Two numbers get confused here and should not be. $\log_2 K$ is the most a source of this size *could* carry. $\lceil \log_2 K \rceil$ is what a fixed-length code *costs*. The first gap is the source being uneven; the second is rounding.

6.3 Extended sources

Nothing forces a coder to work one symbol at a time. Group them in ns and treat each block as one symbol of a new alphabet. That is the **n-th extension**, written S^n, and it has K^n symbols.

> EXTENSION
>
> $$H(S^n) = n\, H(S)$$
>
> The source is memoryless, so the symbols in a block are independent and their information adds. This is the third property of self-information, applied n times.

EXAMPLE 6.2 — THE SAME SOURCE, TWO AT A TIME

GIVEN The source of Example 6.1, extended by two.

FIND $H(S^2)$.

METHOD S^2 has $3^2 = 9$ symbols with probabilities $0.49, 0.14, 0.07, 0.14, 0.04, 0.02, 0.07, 0.02, 0.01$.

SOLUTION Summing $-p \log_2 p$ over all nine gives 2.3136 bits a block.

CHECK $2 \times 1.1568 = 2.3136$. The long way and the short way agree, and per symbol nothing has changed — 2.3136 over two symbols is 1.1568 each. What changes is how much of a whole bit gets wasted in rounding, and section 6.6 turns that into a bound.

ONLY BECAUSE IT IS MEMORYLESS

If the source had memory. As English does, where q is followed by u. The block probabilities would not be products, and $H(S^n)$ would be *less* than $nH(S)$. That difference is exactly the redundancy real compressors live on, and it is outside this course.

6.4 What a code costs

A source encoder turns each symbol into a string of bits, its **codeword**. The codewords need not all be the same length, and the good idea of this chapter is that they should not be. Short ones for common symbols, long ones for rare.

AVERAGE LENGTH AND EFFICIENCY

$$\bar{L} = \sum_{k=1}^{K} p_k\, l_k, \qquad \eta = \frac{L_{\min}}{\bar{L}} = \frac{H(S)}{\bar{L}} \le 1$$

The **source-coding theorem** says $\bar{L} \ge H(S)$ for any code from which the symbols can be recovered exactly. So $L_{\min} = H(S)$: the entropy is not merely a measure of information, it is a limit on how few bits will carry it.

SOURCE CODING IS NOT CHANNEL CODING

Source coding removes bits the message does not need. Channel coding adds bits back so that errors can be found and fixed. They pull in opposite directions, they are done in that order, and this course covers only the first.

Written English gives a concrete case. Its entropy is estimated at about 1.3 bits a letter, while a typical letter-by-letter variable-length code reaches $\bar{L} = 4.22$, so $\eta = 0.31$. Morse code is the same idea by hand: one dot for e, four symbols for z.

6.5 Codes that can be read back

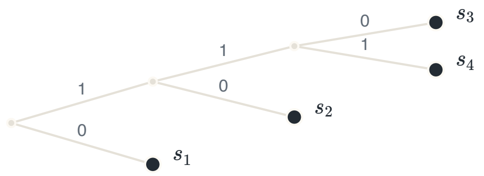
Short codewords are useful only when the receiver can separate them. A code is **uniquely decodable** if each coded bit string has one source-symbol sequence. A **prefix code** has no codeword that begins another codeword. This stronger condition permits immediate decoding when the last codeword bit arrives. A prefix code is also **instantaneous**.

SYMBOL	CODE I	CODE II	CODE III
s_1	0	0	0
s_2	1	10	01
s_3	00	110	011
s_4	11	111	0111

Code I is not a prefix code and is not uniquely decodable either: the string 00 is both s_3 and $s_1 s_1$. Code II is a prefix code. Code III is not a prefix code, since 0 begins all three of the others.

CODE III

Code III is uniquely decodable because each codeword starts with 0 and then contains only 1s. However, it is not instantaneous. After 011, the decoder must wait for the next bit to distinguish s_3 from the start of s_4.



Code II. Every symbol is at a leaf, so no path to one passes through another. That is what the prefix property looks like.

Code III. Every symbol sits on one path, each hanging off the node before it. Nothing is at a leaf but the last, and that is why the decoder must wait.

6.6 The Kraft inequality, and how close a code can get

Before selecting codewords, determine whether their lengths can satisfy the prefix condition. Treat the binary tree as one unit. A codeword of length l uses a fraction 2^{-l} of the tree. These fractions cannot sum to more than one.

KRAFT INEQUALITY

$$\sum_{k=1}^{K} 2^{-l_k} \leq 1$$

The three Kraft sums are 1.5, 1, and 0.9375. Code I fails, so no prefix code has its lengths. Code II uses the tree completely. Code III satisfies the inequality but is not itself a prefix code. Thus, the inequality is necessary but not sufficient for given codewords.

HOW CLOSE A PREFIX CODE GETS

$$H(S) \le \bar{L} < H(S) + 1$$

$$H(S) \le \frac{L_n}{n} < H(S) + \frac{1}{n} \quad \Longrightarrow \quad \lim_{n\to\infty} \frac{L_n}{n} = H(S)$$

The ideal length for symbol k is $-\log_2 p_k$, almost never a whole number. Therefore, each length is rounded up and the rounding costs under one bit. Applying the same bound to the n-th extension spreads that one bit over n symbols.

DYADIC PROBABILITIES

If every probability has the form $p_k = 2^{-l_k}$, the distribution is **dyadic**. No codeword length requires rounding. Then $\bar{L}$ and $H(S)$ equal the same sum. Therefore, $\bar{L} = H(S)$.

BLOCK-CODE SIZE

The coding gap decreases as $1/n$, but the block alphabet grows as K^n. A gap of 0.05 bit per symbol needs $n = 20$. For a three-symbol source, this block contains $3^{20} = 3.5 \times 10^9$ possible messages.

6.7 Huffman coding

The bound says a good code exists. Huffman coding builds one, and it builds the best one: no prefix code on single symbols has a smaller average length.

THE ALGORITHM

1. List the symbols in order of decreasing probability.
2. Take the two least likely, label one 0 and the other 1, and merge them into one symbol whose probability is their sum. Re-sort.
3. Repeat until two symbols remain, and label those 0 and 1.
Each codeword is the labels along the path back to its symbol.

EXAMPLE 6.3 — THE STANDARD FIVE-SYMBOL SOURCE

GIVEN $p = 0.4,\ 0.2,\ 0.2,\ 0.1,\ 0.1.$

MERGING $0.1 + 0.1 = 0.2$. Then the two smallest are 0.2 and 0.2, giving 0.4. Then 0.2 and 0.4 give 0.6. Then 0.4 and 0.6 finish it.

SOLUTION Lengths $2, 2, 2, 3, 3$, so $\bar{L} = 0.4(2) + 0.2(2) + 0.2(2) + 0.1(3) + 0.1(3) = 2.2$ bits a symbol.

EFFICIENCY $H(S) = 2.1219$ bits, so $\eta = 2.1219/2.2 = 0.9645$ — the code spends 3.68% more than the source strictly needs.

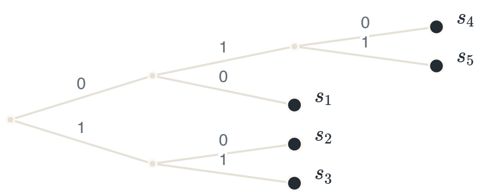
CHECK $2.1219 \le 2.2 < 3.1219$: the two-sided bound holds. A fixed-length code would have cost $\lceil \log_2 5 \rceil = 3$ bits, so Huffman saved 0.8 bits a symbol.

Two Huffman choices are free. Either symbol in a merged pair can receive 0. A merged symbol can also occupy different positions during a probability tie. These choices give different codes with the same minimum average length. Their codeword-length variances can differ.

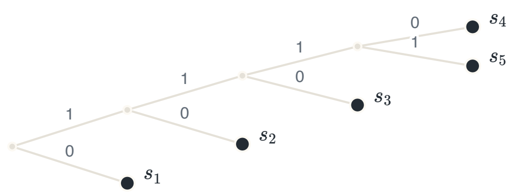
VARIANCE OF THE CODEWORD LENGTH

$$\sigma^2 = \sum_{k=1}^{K} p_k \left(l_k - \bar{L}\right)^2$$

For the example: placing the merged symbol high gives lengths $2, 2, 2, 3, 3$ and $\sigma^2 = 0.16$. Placing it low gives $1, 2, 3, 4, 4$, the same $\bar{L} = 2.2$, and $\sigma^2 = 1.36$.



Merged symbol placed high. Lengths $2, 2, 2, 3, 3$, average 2.2, variance 0.16. The tree is nearly balanced.



Merged symbol placed low. Lengths $1, 2, 3, 4, 4$, the same average 2.2, variance 1.36. The tree is a ladder.

MINIMUM LENGTH VARIANCE

Variable codeword lengths make a buffer fill and empty at changing rates. Smaller length variance reduces these changes. During a probability tie, place the merged symbol as high as possible. This rule gives the minimum-variance Huffman code.

6.8 Lempel–Ziv coding

Huffman coding has one practical weakness, and it is not its length. It needs the probabilities *before* it can build the tree, and for a stream arriving over a wire nobody knows them in advance. A **universal** code is one that reaches the entropy without being told the source, and the Lempel–Ziv algorithm is the one in use everywhere.

THE ALGORITHM

Read the stream once. Each time, take the **shortest run of bits not seen before** and store it in a dictionary. Because it is the shortest new one, everything but its last bit is already stored. So it is sent as a **pointer** to that earlier entry followed by the one new bit, the **innovation**. The decoder builds the same dictionary from the same blocks in the same order. Therefore, the dictionary is never transmitted.

EXAMPLE 6.4 — PARSING AND SENDING A STREAM

GIVEN The stream $0\ 1\ 00\ 011\ 1\ldots$, with 0 and 1 held in the dictionary at positions 1 and 2.

METHOD Parse left to right into pieces not seen before, then send each piece as (position of its start, new bit).

SOLUTION 00 takes position 3: its start 0 is at position 1 and its new bit is 0, so with four-bit blocks it goes as $001\,0$. Next 011 takes position 4: its start 01 is at position 4… so it is sent as $100\,1$.

DECODING Receive 1101. The last bit is the innovation 1. The first three bits, 110, give pointer 6. Therefore, append 1 to dictionary entry 6. The decoder already contains this entry.

CHECK Seven pieces at four bits each cost 28 bits where the raw stream was 18. The method *loses* on a short stream, and that is not a fault: the dictionary has to be paid for before it can pay back. In practice the block is 12 bits, giving $2^{12} = 4096$ entries, and a long file compresses to roughly two thirds of its size.

OPTIMAL AGAINST UNIVERSAL

Huffman is optimal but not universal: no prefix code on single symbols beats it, and it cannot start until the probabilities are known. Lempel–Ziv is universal but reaches the limit only in the long run. That trade is the whole difference between them, and it is why the second is what compresses an archive.

6.9 The discrete memoryless channel

The first part of this chapter studied the source. The second part studies the channel. A **discrete memoryless channel** maps one symbol from a finite input alphabet to a finite output alphabet. The current output depends only on the current input.

THE CHANNEL, AND WHAT THE TRANSMITTER ADDS

$$p(y_k \mid x_j) = P\big(Y = y_k \mid X = x_j\big), \qquad \sum_{k=0}^{K-1} p(y_k \mid x_j) = 1 \ \text{ for every } j$$

$$p(x_j, y_k) = p(y_k \mid x_j)\, p(x_j), \qquad p(y_k) = \sum_{j=0}^{J-1} p(y_k \mid x_j)\, p(x_j)$$

The transition probabilities collected into a matrix, one row per input, are the **channel matrix**. Every *row* sums to one, because something must come out when a symbol goes in. The columns do not, and expecting them to is the usual first mistake. The **input distribution** $p(x_j)$ is not part of the channel — it belongs to the transmitter, and it is chosen.

THE BINARY SYMMETRIC CHANNEL

Two inputs, two outputs, and one number. The probability p that a bit is flipped, called the **crossover probability**. Its matrix is $\begin{bmatrix} 1-p & p \\ p & 1-p \end{bmatrix}$. It is not an abstraction invented for this chapter. The binary receiver of chapter 2, seen from outside, *is* a binary symmetric channel with $p = Q\left(\sqrt{2E_b/N_0}\right)$. With equally likely inputs the output is equally likely whatever p is, even at $p = \frac{1}{2}$ where the channel is destroying everything. Therefore, a balanced output is no evidence that anything survived.

6.10 Mutual information

Before anything arrives, the uncertainty about the input is $H(X)$. Then one output symbol arrives, and it rarely settles the question but it does change the odds. What remains is an entropy like any other, averaged over which output actually came.

CONDITIONAL ENTROPY, AND THE INFORMATION THAT GOT THROUGH

$$H(X \mid Y) = \sum_k H(X \mid Y = y_k)\, p(y_k) = -\sum_k \sum_j p(x_j, y_k) \log_2 p(x_j \mid y_k)$$

$$I(X;Y) = H(X) - H(X \mid Y) = H(Y) - H(Y \mid X)$$

$$H(X,Y) = H(X) + H(Y) - I(X;Y)$$

Read the middle line in words: **how much doubt the arrival removed**, in bits, per use of the channel. The second form is the one used, because $H(Y \mid X)$ comes straight off the channel matrix while $H(X \mid Y)$ needs Bayes' rule first.

THREE PROPERTIES

Symmetry. $I(X;Y) = I(Y;X)$. Bayes' rule gives $p(x_j \mid y_k)/p(x_j) = p(y_k \mid x_j)/p(y_k)$, and substituting turns one sum into the other. What the output says about the input equals what the input says about the output — which is why the quantity is called *mutual*.

Non-negativity. $I(X;Y) \geq 0$, with equality exactly when X and Y are independent. An observation never leaves you knowing less than before.

Joint entropy. Adding $H(X)$ and $H(Y)$ counts the shared part twice, so it is subtracted once. That is the whole content of the third line.

EXAMPLE 6.5 — THE BINARY SYMMETRIC CHANNEL AT $p = 0.1$

GIVEN A BSC with $p = 0.1$ and equally likely inputs.

FIND $H(X)$, $H(Y \mid X)$, $H(Y)$ and $I(X;Y)$.

METHOD $H(Y \mid X)$ comes off the matrix: whichever symbol goes in, the output distribution is $(0.9, 0.1)$ in some order. $H(Y)$ needs the output distribution first.

SOLUTION $H(X) = 1$ bit. $H(Y \mid X) = H(0.1) = -0.1 \log_2 0.1 - 0.9 \log_2 0.9 = 0.4690$ bits. The output is equally likely, so $H(Y) = 1$ bit, and $I(X;Y) = 1 - 0.4690 = 0.5310$ bits per use.

CHECK One bit was offered and just over half a bit arrived. The missing 0.4690 bits were not delayed or corrupted. They were destroyed, and no receiver however clever recovers them. The joint entropy is $1 + 1 - 0.5310 = 1.4690$ bits, and at $p = 0.25$ the shared part falls to 0.1887 while the total rises to 1.8113. A noisier channel does not destroy uncertainty, it moves it out of the shared part.

THE TWO CONDITIONAL ENTROPIES ARE NOT THE SAME THING

$H(Y \mid X)$ is the uncertainty about the *output* given the input, and for the BSC it is $H(p)$ whatever the transmitter does. A property of the channel alone. $H(X \mid Y)$ is the uncertainty about the *input* given the output, and it depends on the input distribution too. $I(X;Y) = I(Y;X)$ is a statement about one number. It does not make the two ends interchangeable.

6.11 Channel capacity

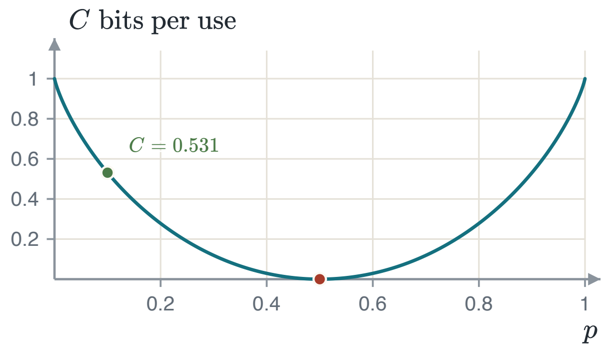
Mutual information measures one transmitter on one channel. Change how often each symbol is sent and the number moves. The channel is fixed and the input distribution is the engineer's to choose. Therefore, take the best choice. What comes out then depends on the channel matrix alone. This occurs because the one free thing has been optimised away. **capacity is a property of the channel, and mutual information is not.**

CAPACITY, AND THE CAPACITY OF THE BSC

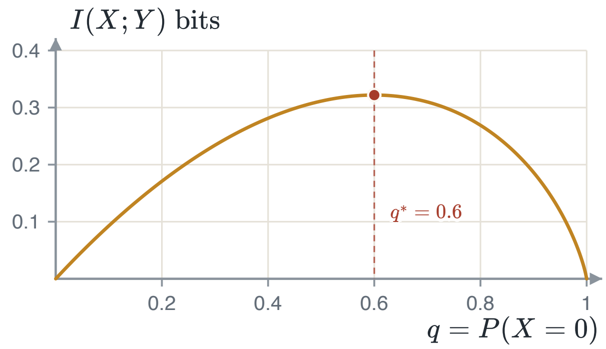
$$C = \max_{\{p(x_j)\}} I(X;Y) \quad \text{bits per channel use}$$

$$C_{\text{BSC}} = 1 - H(p), \qquad H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$$

For the BSC, $H(Y \mid X) = H(p)$ whatever the transmitter does, and $H(Y) \le 1$ bit with equality when the output is equally likely — which equally likely inputs deliver. So the maximum sits at $p(x_0) = \frac{1}{2}$ and the two pieces subtract. The unit is bits per *use* of the channel, not bits per second. Multiply by the number of uses a second to reach a rate.



Capacity of the binary symmetric channel. The capacity is one bit at $p = 0$ and $p = 1$. At $p = 1$, the receiver restores every bit by inversion. Capacity is zero at $p = \frac{1}{2}$, where the output is independent of the input.



The Z-channel of Example 6.6. Its mutual information peaks at $q = 0.6$, not at a half, which is what an asymmetric channel does: the transmitter should favour the symbol that survives.

The curve is flat near $p = 0$: a channel with one error in a thousand has capacity 0.9886 bits, so the first errors cost almost nothing. It then falls away steeply — at $p = 0.11$ the capacity is already 0.500. Halving the error probability of an already-good channel buys very little, and the effort belongs where the curve is steep.

EXAMPLE 6.6 — THE CAPACITY OF THE Z-CHANNEL

GIVEN A 0 always arrives as a 0. A 1 arrives correctly half the time and as a 0 otherwise. Write $P(X = 0) = q$.

FIND The capacity, and the input distribution that achieves it.

METHOD Nothing is symmetric, so the equally likely input is no longer the right guess. Form $I(X;Y)$ as a function of q and maximise it.

SOLUTION $P(Y = 0) = q + \frac{1}{2}(1 - q) = \frac{1}{2}(1 + q)$, and $H(Y \mid X) = q(0) + (1 - q)(1) = 1 - q$ because a transmitted 0 leaves nothing in doubt. Subtracting gives $I(X;Y) = q - \frac{1}{2}(1 + q)\log_2(1 + q) - \frac{1}{2}(1 - q)\log_2(1 - q)$. Setting the derivative to zero collapses the logarithmic terms to $\frac{1}{2}\log_2\frac{1-q}{1+q} = -1$, so $\frac{1-q}{1+q} = \frac{1}{4}$ and $q^* = 0.6$.

CHECK $C = 0.3219$ bits per channel use, and $0.3219 = \log_2\frac{5}{4}$ exactly — a check on the arithmetic that costs nothing. The optimum is not the balanced input: the transmitter should send the reliable symbol three times in five. A transmitter that guesses $q = 0.5$ still gets 0.3113 bits, losing about 3%.

THE CHANNEL CODING THEOREM

If the transmission rate satisfies $R_b < C$, there is a coding scheme whose probability of error is as small as required. If $R_b > C$, there is not — no scheme, however long or however clever. **An error-free data rate can never exceed the capacity.** The theorem says a code exists. It does not say what the code is, how long its blocks must be, or what the decoder costs. Finding codes that come close and can also be decoded took the fifty years after it was proved.

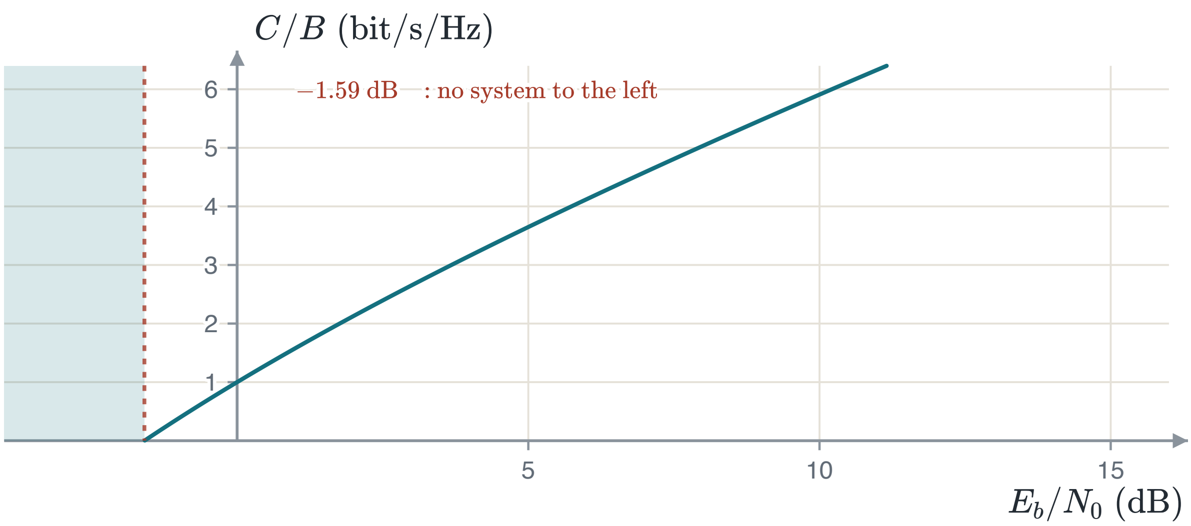
6.12 The bandlimited channel

Everything above counted symbols. A real channel is given instead as a bandwidth in hertz, an average transmitted power and a noise density. Its capacity is one of the most quoted results in engineering.

> THE INFORMATION CAPACITY LAW
>
> $$C = B\log_2\left(1 + \frac{P}{N_0 B}\right) \quad \text{bits per second}$$
>
> $$\frac{C}{B} = \log_2\left(1 + \frac{E_b}{N_0}\frac{C}{B}\right) \quad \Longrightarrow \quad \frac{E_b}{N_0} = \frac{2^{C/B}-1}{C/B}$$
>
> B is bandwidth, P is average transmitted power, and $N_0 B$ is in-band noise power. Use $P = E_b C$, where power equals energy per bit times bit rate. Divide by B to obtain the bandwidth efficiency C/B.

> READ THE TWO KNOBS
>
> C grows **linearly** with bandwidth and only **logarithmically** with power. Doubling the bandwidth at fixed noise density roughly doubles the rate. Doubling the power adds one bit per second per hertz at best and much less when the ratio is already large. Bandwidth is the better buy, and it is the one that is scarce and regulated. At $C/B = 1$ the requirement is $E_b/N_0 = 1$, which is 0 dB. At $C/B = 2$ it is 1.5, which is 1.76 dB.

> THE SHANNON LIMIT
>
> $$\lim_{x\to 0}\frac{2^x - 1}{x} = \ln 2, \qquad \left.\frac{E_b}{N_0}\right|_{\min} = \ln 2 = 0.693 = -1.59\ \text{dB}$$
>
> Spend bandwidth freely — let $C/B \to 0$ — and the energy per bit a system needs falls, but not to nothing. No system of any kind communicates reliably below -1.59 dB: not a better code, not a better modulation, not a better receiver.



Bandwidth efficiency against energy per bit. Every working system sits below the curve and to the right of the line. The region to the left of -1.59 dB has never been occupied and never will be. Climbing from one bit per hertz to six costs about 14 dB, and the climb steepens the whole way.

A FLOOR THAT IS APPROACHED AND NEVER TOUCHED

Reaching the limit needs infinite bandwidth, and the rate per hertz goes to zero on the way. Coherent binary PSK needs about 9.6 dB for an error probability of 10^{-5}, so it sits some 11 dB above the floor. Closing that gap is what channel coding was invented for. This course stops at the uncoded schemes, which is where the gap is widest and easiest to see.

6.13 Summary

RESULT	STATEMENT	ANCHOR
Self-information	$I(s_k) = -\log_2 p_k$	PS CH12.1.1
Entropy	$H(S) = -\sum_k p_k \log_2 p_k$	PS CH12.1.1
Bounds	$0 \le H(S) \le \log_2 K$	PS CH12.1.1
Extension	$H(S^n) = nH(S)$	PS CH12.1.1
Average length	$\bar{L} = \sum_k p_k l_k$, $\eta = H(S)/\bar{L}$	PS CH12.2
Source-coding theorem	$\bar{L} \ge H(S)$, so $L_{\min} = H(S)$	PS CH12.2
Kraft	$\sum_k 2^{-l_k} \le 1$: necessary, not sufficient	PS CH12.3
Prefix bound	$H(S) \le \bar{L} < H(S) + 1$	PS CH12.2
Huffman	optimal, not unique. High placement gives least variance	PS CH12.3.1

Chapter 1 turned a waveform into bits. Chapters 2 to 5 got those bits across a channel and counted the errors. This chapter asked how few bits there needed to be in the first place, and answered with one number that the source itself decides. Everything between the two is engineering; the entropy is not.

APPENDIX A

Summary of formulas

Everything the course establishes, in the order it establishes it. Nothing here is derived — where a result needs an argument, the argument is in the chapter named beside it.

A.1 From an analog signal to bits — Chapter 1

SAMPLING AND RECONSTRUCTION

$$f_s > 2W \quad \text{(the sampling theorem, } W \text{ the highest frequency present)}$$

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT_s)\,\operatorname{sinc}\left(\frac{t - nT_s}{T_s}\right)$$

Below the Nyquist rate the replicas of the spectrum overlap and the overlap cannot be undone. $\operatorname{sinc}(x) = \sin(\pi x)/(\pi x)$ throughout this course.

UNIFORM QUANTIZATION

$$\Delta = \frac{2V}{L} = \frac{2V}{2^n}, \qquad |e| \le \frac{\Delta}{2}$$

$$\sigma_q^2 = \frac{\Delta^2}{12}, \qquad \mathrm{SQNR}\big|_{\mathrm{dB}} = 6.02n + 1.76$$

The 1.76 assumes a full-scale sinusoid. The formula is an approximation at small n. At three bits it gives 19.82 dB where the measured value is 19.09 dB, and the gap halves with each extra bit.

RESULT	STATEMENT	CHAPTER
Companding	μ-law and A-law compress before quantizing so that the signal-to-noise ratio is flat across the range	1
PCM rate	$R = nf_s$ bits a second for n bits a sample	1

A.2 Baseband transmission — Chapter 2

THE MATCHED FILTER

$$h(t) = s(T-t), \qquad \left(\frac{S}{N}\right)_{\max} = \frac{2E}{N_0}$$

The matched filter maximizes the signal-to-noise ratio at the sampling instant. The maximum depends on pulse energy, not pulse shape.

BINARY ERROR PROBABILITY AT BASEBAND

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) \text{ (antipodal)}, \qquad P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right) \text{ (on-off, orthogonal)}$$

NYQUIST CRITERION FOR ZERO INTERSYMBOL INTERFERENCE

$$\sum_k P\left(f + \frac{k}{T}\right) = T \quad \text{for all } f$$

$$B = \frac{1+\alpha}{2T}, \qquad 0 \le \alpha \le 1$$

The raised cosine meets the criterion for every roll-off α. At $\alpha = 0$ the bandwidth is the Nyquist minimum $1/2T$ and the pulse decays slowly. At $\alpha = 1$ it is twice that and the pulse decays quickly.

A.3 Signal space — Chapter 3

A SIGNAL AS A POINT

$$s_i(t) = \sum_{k=1}^{N} s_{ik}\psi_k(t), \qquad s_{ik} = \int_0^T s_i(t)\psi_k(t)\,dt$$

$$E_i = \|\mathbf{s}_i\|^2 = \sum_{k=1}^{N} s_{ik}^2, \qquad d_{ij} = \|\mathbf{s}_i - \mathbf{s}_j\|$$

The basis $\{\psi_k\}$ is orthonormal and comes from the Gram–Schmidt procedure applied to the signal set. Energy becomes a squared length and difference becomes a distance, which is the move the rest of the course rests on.

A.4 The optimal receiver — Chapter 4

THE DECISION RULE

$$\hat{s} = \arg\min_i \left\{ \|\mathbf{r} - \mathbf{s}_i\|^2 - N_0 \ln P(\mathbf{s}_i) \right\} \quad \text{(MAP)}$$

$$\hat{s} = \arg\min_i \|\mathbf{r} - \mathbf{s}_i\|^2 \quad \text{(ML, equal priors)}$$

With equal priors the rule is minimum distance. The equivalent correlation form is to maximise $\mathbf{r} \cdot \mathbf{s}_i - E_i/2$.

ERROR PROBABILITY

$$P(\mathbf{s}_k \to \mathbf{s}_j) = Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right)$$

$$P_e \le \frac{1}{M} \sum_k \sum_{j \ne k} Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right), \qquad P_e \approx N_{\min}\, Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right)$$

The first result is exact for two points. The second is the union bound. The third is its nearest-neighbor form. Here, $N_{\min}$ is the average number of nearest neighbors and can be noninteger. The loose bound needs only M and $d_{\min}$.

A.5 Digital modulation — Chapter 5

SCHEME	$d^2_{\min}$	$N_{\min}$
BPSK	$4E_b$	1
BFSK, BASK	$2E_b$	1
M-PSK	$4E_s \sin^2(\pi/M)$	2
M-PAM	$12E_s/(M^2-1)$	$2(M-1)/M$
M-QAM (square)	$6E_s/(M-1)$	3 at $M=16$, 3.5 at $M=64$
M-FSK	$2E_s$ for every pair	$M-1$

READING THE TABLE

$$P_e \approx N_{\min}\, Q\left(\sqrt{\frac{d^2_{\min}}{2N_0}}\right), \qquad E_s = (\log_2 M)E_b$$

Convert once between symbol and bit energy. BFSK and BASK need 3.01 dB more than BPSK for the same error probability. Doubling M costs approximately 6 dB in PAM. Moving from 4-PSK to 8-PSK costs 5.33 dB. At the same energy, 16-QAM gains 9.29 dB over 16-PAM.

A.6 Information theory — Chapter 6

INFORMATION AND ENTROPY

$$I(s_k) = -\log_2 p_k, \qquad H(S) = -\sum_{k=1}^{K} p_k \log_2 p_k$$

$$0 \le H(S) \le \log_2 K, \qquad H(S^n) = n\, H(S)$$

The upper bound is reached when all K symbols are equally likely. The extension result holds because the source is memoryless.

SOURCE CODING

$$\bar{L} = \sum_{k=1}^{K} p_k l_k, \qquad \eta = \frac{H(S)}{\bar{L}} \le 1$$

$$\sum_{k=1}^{K} 2^{-l_k} \le 1, \qquad H(S) \le \bar{L} < H(S) + 1$$

The Kraft inequality is necessary but not sufficient for a prefix code. Over the n-th extension the upper bound becomes $H(S) + 1/n$, so $L_n/n \to H(S)$.

HUFFMAN CODES

$$\sigma^2 = \sum_{k=1}^{K} p_k \left(l_k - \bar{L}\right)^2$$

Huffman is optimal in average length and is not unique. Placing a merged symbol as high as possible on a tie gives the minimum-variance code among the optimal ones.

APPENDIX B

The laboratories

The course has four laboratories, and each one takes a result the chapters derive on paper and asks you to measure it instead. This appendix runs all four. Every curve, cloud and staircase on the pages that follow was computed when this page was drawn. The noise was generated, the filter was applied, the errors were counted. So what you see is the outcome of a run and not a drawing of one.

That matters for one reason above all. A formula tells you what should happen. A measurement tells you what did. Where the two agree you have learnt that the derivation is sound. Where they disagree you have learnt something better: which assumption in the derivation the experiment broke. Each of the four sections below ends on that disagreement, because it is the part of the laboratory worth carrying away.

Your numerical results will differ from these examples. Each group receives a different input number, so its waveform, bit pattern, and source statistics differ. Random noise also changes between runs. Compare curve shapes, gaps, trends, and expected directions instead of individual sample values.

LABORATORY SUMMARY

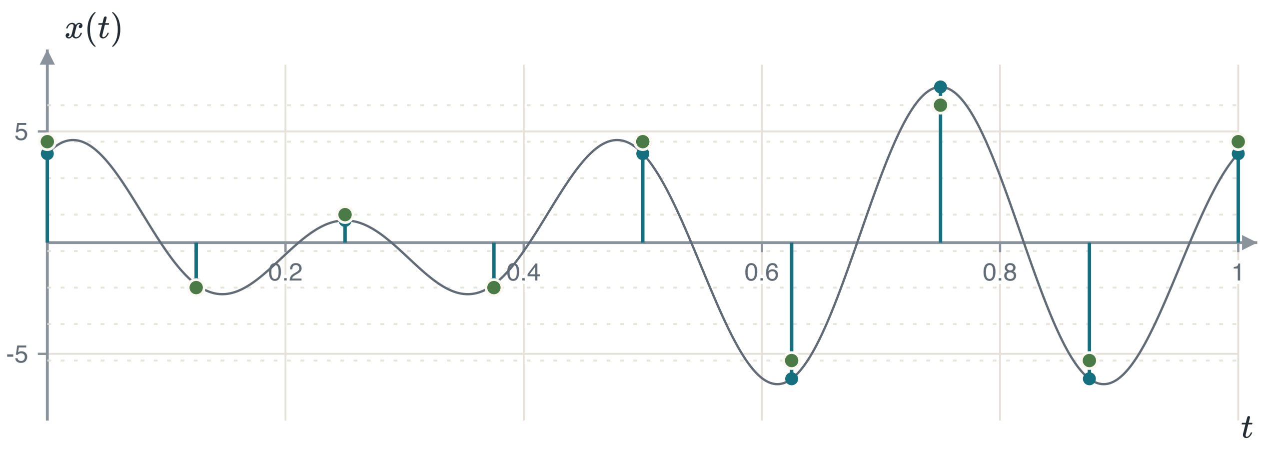
B.1 builds a uniform quantizer, measures SQNR, and encodes the samples as a polar-NRZ waveform. **B.2** measures the bit error rate of a matched-filter receiver and compares it with the Chapter 2 formula. **B.3** measures the symbol error rate of sixteen-point QAM and compares it with the Chapter 4 bound. **B.4** builds a Huffman code and measures its compression.

B.1 Quantization, the signal-to-quantization-noise ratio and PCM

The first laboratory applies the complete Chapter 1 chain. Start with a continuous waveform and sample it at the Nyquist rate. Quantize the samples with a uniform quantizer. Measure the ratio of signal power to quantization-error power. Encode each sample as a binary word and draw the polar-NRZ waveform. Then double the number of levels and repeat the measurements.

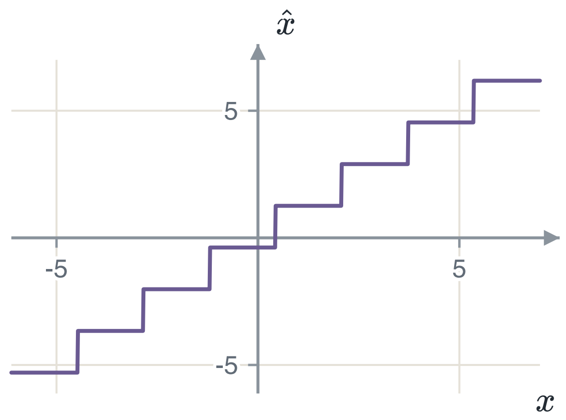
The waveform is $x(t) = f_1 \sin(2\pi f_1 t) + f_2 \cos(2\pi f_2 t)$. Each group receives integer values for f_1 and f_2. Here, $f_1 = 3$ and $f_2 = 4$, so the Nyquist rate is 8 samples per second. A two-second interval then contains only seventeen samples.

The quantizer itself is four lines. Find the smallest and largest sample. Divide the distance between them into L equal cells. Therefore, the step is $\Delta = (x_{\text{max}} - x_{\text{min}})/L$. Decide which cell each sample falls in. Replace it by the middle of that cell. This is the value that minimises the error inside the cell when nothing is known about where in the cell the sample sits.

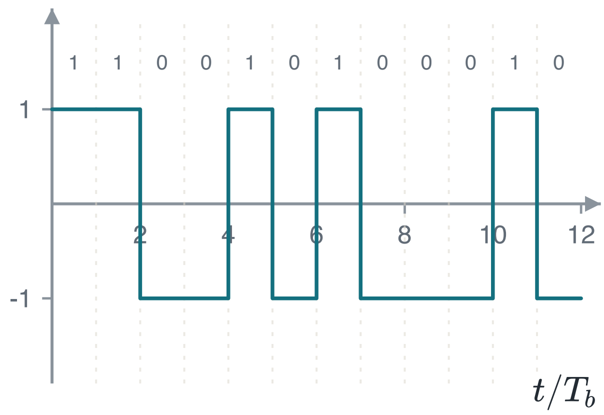


The first second of the run. The pale line is $x(t)$. Stems show samples x_n taken every $1/8$ s. Filled circles show quantized values $\hat{x}_n$ for $L = 8$. Dotted lines show the eight representation levels with spacing $\Delta = 1.640$ V.

With $L = 8$ each sample needs $R = \log_2 8 = 3$ bits, and at $f_s = 8$ samples a second the stream leaves the encoder at $R_b = Rf_s = 24$ bit/s. At $L = 16$ it is 32 bit/s. That is the price of the second experiment, and it is worth naming before the reward is measured: a third more bits a second.



The quantizer itself, drawn by feeding it a thousand values across its range. Eight treads, each $\Delta = 1.640$ V wide, each one tread high.



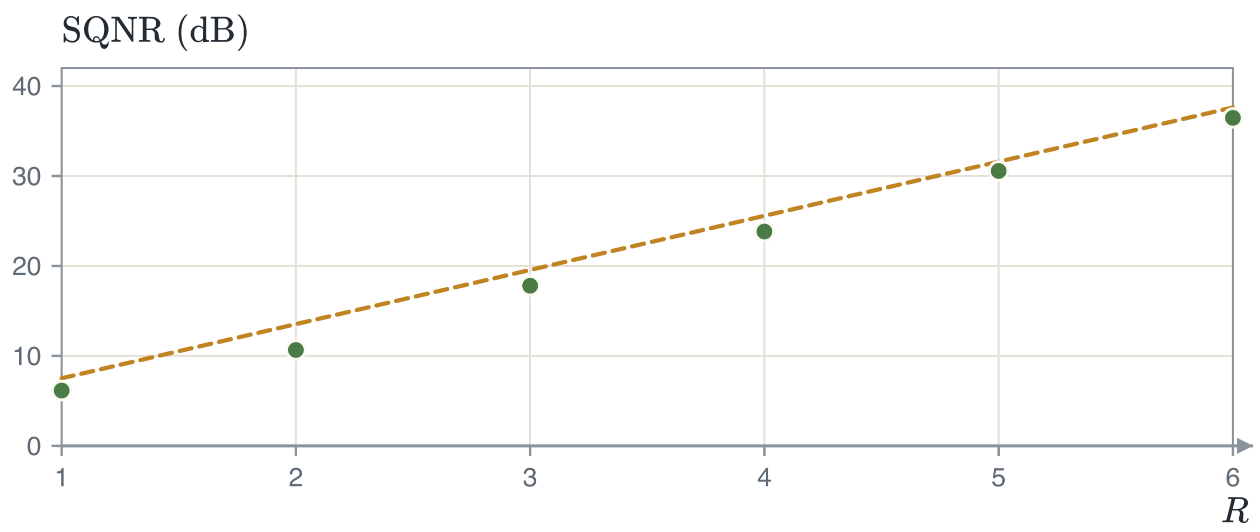
The first four samples encoded and sent. Each sample becomes a three-bit word for its cell, and each bit becomes one bit interval of a polar NRZ waveform. High for a one, low for a zero.

WHAT THE LABORATORY MEASURES

$$\mathrm{SQNR} = \frac{E[X^2]}{E\big[(X - \hat{X})^2\big]} \longrightarrow \mathrm{SQNR}\big|_{\mathrm{dB}} = 10\log_{10}\mathrm{SQNR}$$

Both averages use the available samples. The numerator is the sample mean-square value. The denominator is the mean-square quantization error. Chapter 1 models this denominator as $\Delta^2/12$. Here, it is measured directly.

On this waveform the measurement gives 17.80 dB at $L = 8$ and 23.83 dB at $L = 16$. The difference is 6.03 dB. This is the six decibels a bit that Section 1.4 predicts, arrived at by counting. That agreement is the main result of the laboratory, and it holds for a reason that has nothing to do with this particular waveform. Doubling L halves Δ, halving Δ quarters $\Delta^2/12$, and a factor of four is 6.02 dB.



Measured against modelled, for one to six bits a sample, both computed from the seventeen samples the laboratory takes. The dashed line is the small-step model $10\log_{10}\left(P_X/(\Delta^2/12)\right)$ and the circles are the ratio actually measured. The circles rise about six decibels a bit, as they should. But they sit below the line by between 1.0 and 2.9 dB, and the gap neither closes nor settles as R grows. Something is wrong, and the next note says what.

SEVENTEEN SAMPLES IS NOT A POPULATION

The measured points and the modelled line disagree by up to 2.9 dB, and the gap does not shrink as L grows. It is tempting to blame the model, and Section 1.4 gives a real reason it could be blamed. The error is not quite uniform across a cell. But measure the same waveform with two thousand samples instead of seventeen and the gap falls to less than 0.6 dB everywhere. The model was right. The estimate was noisy. Both $E[X^2]$ and $E[(X-\hat{X})^2]$ are averages, and an average over seventeen numbers has a spread of its own. This is the error to name in your report. Do not draw a conclusion about a model from a sample too small to test it.

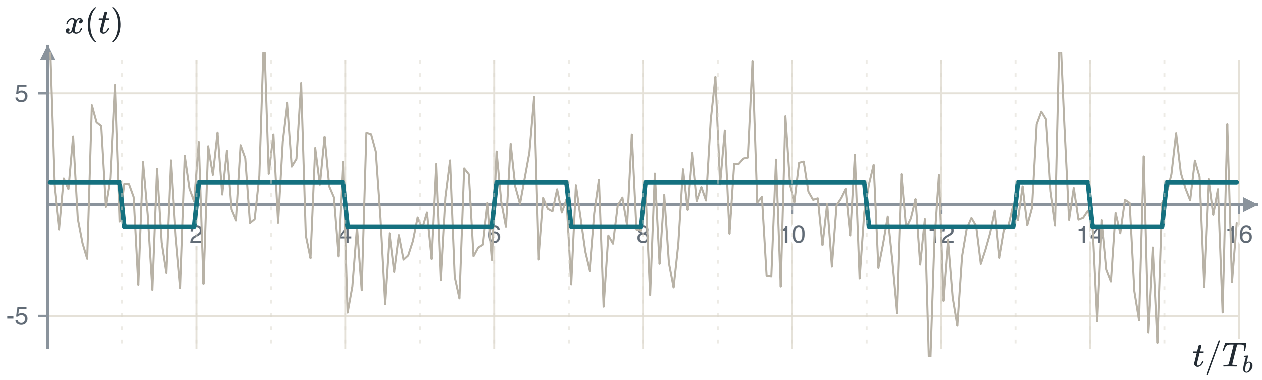
SAMPLING EXACTLY AT THE NYQUIST RATE

The waveform here has a 4 Hz component and the laboratory samples at exactly 8 Hz. This is the boundary the sampling theorem is stated at rather than inside. At exactly twice its frequency, a cosine is sampled at the same two points of every cycle. Therefore, that component contributes the same alternating ± 4 to every sample and none of its shape survives. The reconstruction is still correct here because the component is a cosine and lands on its peaks. Move the phase and it would not be. Chapter 1 asks for $f_s > 2W$ and means the strict inequality.

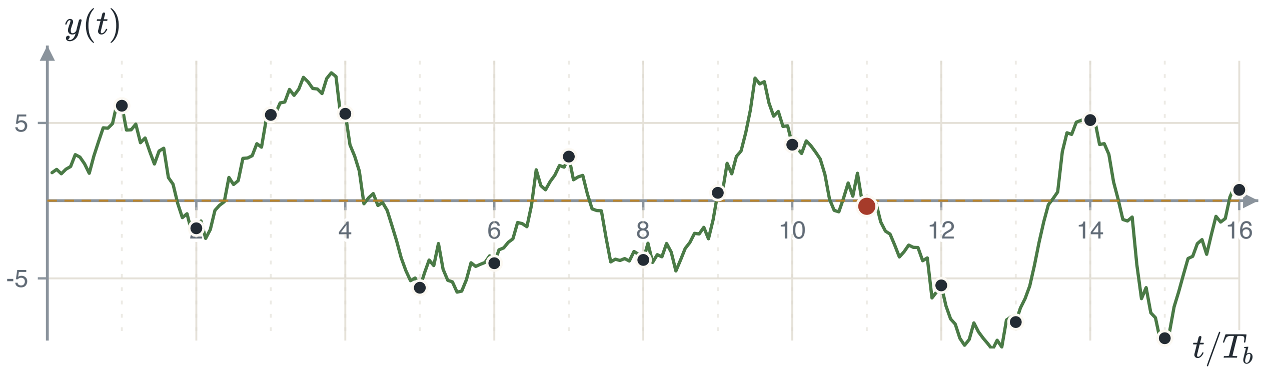
B.2 The matched filter and the bit error rate

The second laboratory builds the Chapter 2 receiver. Convert a bit stream into a waveform and add white Gaussian noise. Apply the matched filter and sample its output once per bit. Compare each sample with the decision threshold. Then compare the measured error rate with $Q\left(\sqrt{2E_b/N_0}\right)$.

The signaling is antipodal. The pulse $\psi(t) = 1/\sqrt{T_b}$ has unit energy over one bit interval. A bit scales this pulse by $\pm A\sqrt{T_b}$. Thus, the transmitted waveform is constant at $+A$ or $-A$ during each interval. Its bit energy is $E_b = A^2 T_b$. Add sample noise with variance $N_0/2$ and use $10\log_{10}(E_b/N_0)$.

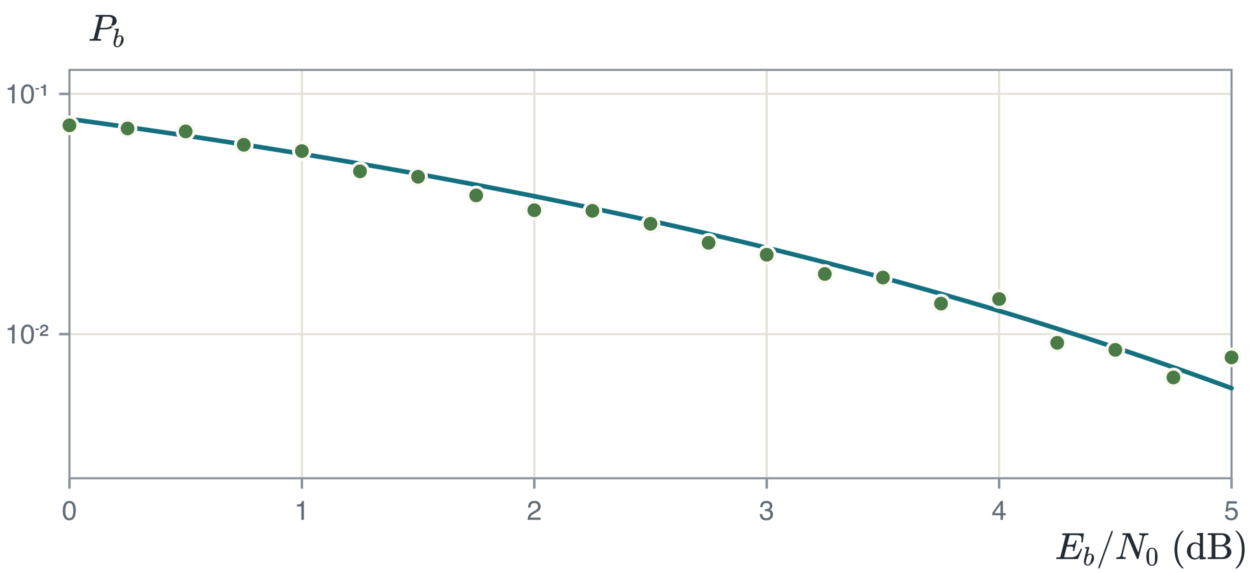


Sixteen bits at $E_b/N_0 = 2$ dB. The clean square wave is the transmitted $s(t)$. The hairline around it is $x(t) = s(t) + w(t)$. This is everything the receiver is given. Nothing about the square wave is visible in the hairline at this noise level, and that is the honest picture. The receiver does not see the signal, it computes with it.



The matched-filter output for the same sixteen bits, read once at the end of every bit interval. The dashed line is the threshold $\lambda = 0$. This is where equal priors put it. Fifteen of the sixteen readings are on the correct side of it. The eleventh is not, and it is drawn in red. At 2 dB the formula expects about one wrong bit in twenty-seven, and this run produced one in sixteen.

One run of sixteen bits settles nothing. One error could as easily have been none or three. The second part of the laboratory therefore drops the picture and repeats the experiment. Five thousand bits at each signal-to-noise ratio from 0 to 5 dB in quarter-decibel steps, counting errors and dividing by five thousand.



Measured bit error rate against the formula, five thousand bits a point. The line is $Q\left(\sqrt{2E_b/N_0}\right)$ and the circles are counted errors. Nowhere on the sweep do the two differ by more than 0.005, and yet the circles clearly wander further from the line towards the right. The next note is about why those two statements are both true.

MONTE CARLO SCATTER

Counting k errors in N bits gives a standard spread of approximately $\sqrt{p(1-p)/N}$. At 0 dB, $p \approx 0.079$, and five thousand bits give a spread of 0.004. At 5 dB, the absolute spread decreases, but it becomes a larger fraction of p. A logarithmic plot shows this relative scatter. Measuring probabilities near 10^{-6} therefore requires millions of trials.

THREE-DECIBEL CONVENTION ERRORS

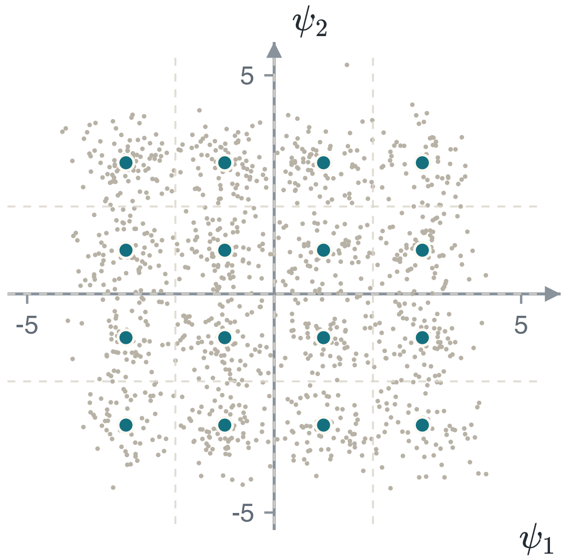
Two convention errors each give a factor of two. First, the sample-noise variance is $N_0/2$ for a two-sided density $N_0/2$. Using N_0 shifts the measured curve by 3 dB. Second, the formula $Q\left(\sqrt{2E_b/N_0}\right)$ requires antipodal signaling. On-off signaling instead uses $Q\left(\sqrt{E_b/N_0}\right)$. A parallel 3 dB shift usually indicates one of these errors.

B.3 Sixteen-point QAM and the union bound

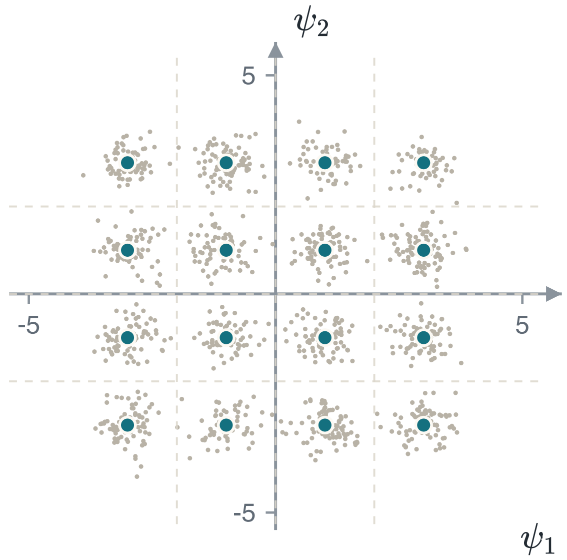
The third laboratory uses a two-dimensional sixteen-point square constellation. Add independent noise on each axis. The receiver selects the nearest point. Compare the measured symbol error rate with the nearest-neighbor approximation from Section 4.4.

Write each point as $s_I + js_Q$, with coordinates from $\{-3, -1, 1, 3\}$. These units give $E_s = 10$ and $d_{\min} = 2$. Each symbol carries four bits, so $E_b = E_s/4 = 2.5$. Section 5.4 gives $d_{\min}^2 = 6E_s/(M-1) = 4$. This result agrees with the point spacing.

The same constellation gives the optimal thresholds. Independent axes and equal level spacing produce three vertical and three horizontal boundaries. Each axis uses $\lambda \in \{-2, 0, 2\}$. Thus, nearest-point detection reduces to nearest-level detection on each axis.



$E_b/N_0 = 6$ dB. The clouds touch, so a large fraction of the symbols land across a boundary. The measured symbol error rate here is 0.108: about one symbol in nine is decided wrongly.



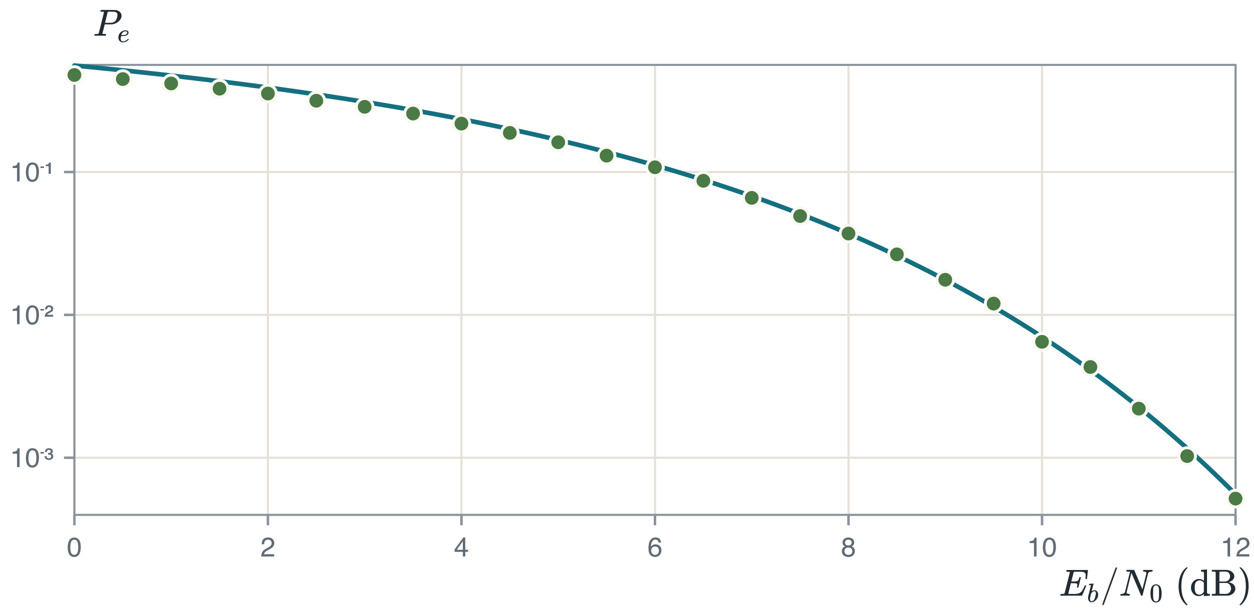
$E_b/N_0 = 11$ dB. Five decibels more energy a bit, and the clouds have pulled apart into sixteen separate blobs. The measured rate falls to 0.0022, about fifty times better.

Twelve hundred received symbols are drawn in each picture so that the clouds can be seen. The rates quoted beneath them do not come from those twelve hundred. A rate of 0.0022 measured on twelve hundred symbols would be three errors, and three is not a measurement. They come from the sweep below. This uses fifty thousand symbols at 6 dB and a quarter of a million at 11 dB. Choosing how many symbols to run is part of the experiment, and Section B.2 said why.

WHAT THE BOUND PREDICTS

$$P_e \;\approx\; N_{\min}\, Q\!\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right), \qquad N_{\min} = 3, \quad d_{\min}^2 = 4$$

$N_{\min}$ is the average number of nearest neighbours: four corner points have two each, eight edge points three, and four inner points four, and $(4 \cdot 2 + 8 \cdot 3 + 4 \cdot 4)/16 = 3$. It is an average over the constellation and it does not have to be a whole number.



Measured symbol error rate against the nearest-neighbour approximation. The number of symbols is raised as the rate falls. Fifty thousand at the left, four hundred thousand at the right. So that every circle rests on a few hundred errors. The circles sit below the line at the left and join it by about 8 dB. That gap is not an error in either the simulation or the formula. It is what the approximation is.

MEASURED ERROR AND THE UNION BOUND

The union bound adds all pairwise error probabilities. These events overlap, so the sum can count one noise vector more than once. The bound is therefore larger than the exact union probability. At low signal-to-noise ratio, overlap makes the bound loose. At high ratios, nearest-neighbor errors dominate and overlap becomes small. Remaining differences between simulated points and the bound can come from finite trial counts.

SYMBOL AND BIT ENERGY

For sixteen-point QAM, $E_s = 4E_b$. Convert between these energies once when calculating N_0. Do not calculate N_0 from one energy and use the other in the Q function. This error shifts the complete curve by $10\log_{10} 4 = 6.02$ dB.

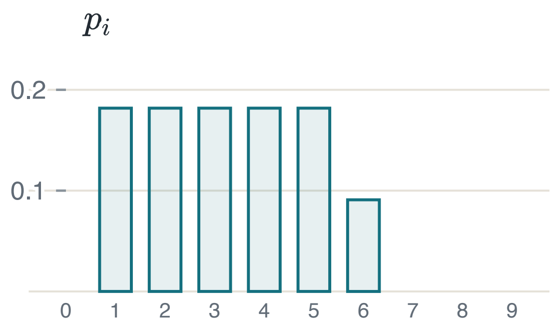
B.4 Huffman coding and how much it compresses

The fourth laboratory studies lossless source coding. A source emits symbols with known probabilities. Entropy gives the minimum average bits per symbol. Huffman coding gives the best single-symbol prefix code. The laboratory measures both values.

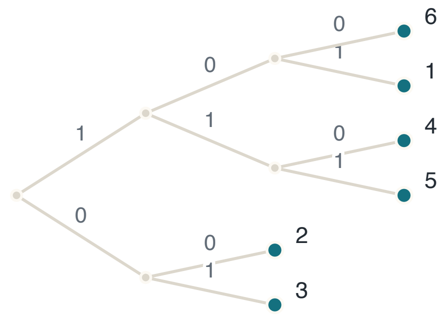
The source alphabet is the ten digits. Your group counts how often each digit occurs in a number it is given and divides by how many digits there are. This gives a probability for each. The case run here uses the eleven digits $1\,2\,3\,4\,5\,6\,5\,4\,3\,2\,1$. Five digits occur twice and one occurs once. Therefore, five symbols have probability $2/11$, one has $1/11$, and four have probability zero and take no codeword at all.

DIVIDING BY THE WRONG TOTAL

The probabilities must come from dividing each count by the *number of symbols*. Dividing by anything else. The sum of the digit values, say. Gives a set of numbers that does not add up to one, and everything downstream is then quietly wrong. The entropy is not an entropy, the codebook is built for a source that does not exist, and the efficiency can come out above 100%. Print the sum of your probability vector and confirm it is one before you compute anything from it. It is one line, and it catches the only mistake in this laboratory that produces plausible numbers instead of an error message.



The source statistics p_i over the ten-digit alphabet. Four digits do not occur, and a symbol of probability zero is left out of both the entropy and the codebook.



The code the algorithm builds. Reading a path from the root, left is a 0 and right a 1. The digit at each leaf is the symbol that path spells.

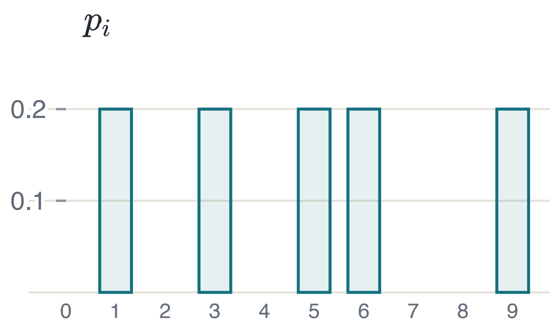
This source has entropy $H = 2.550$ bits per symbol. Its Huffman code has average length $\bar{L} = 2.636$, so $\eta = 96.7\%$. First, check the Section 6.6 bound $H \leq \bar{L} < H + 1$. Here, $2.550 \leq 2.636 < 3.550$. Second, compare $\bar{L}$ with the three-bit fixed length for six symbols and the four-bit length for ten symbols.

A TIE IN THE ALGORITHM IS NOT A MISTAKE

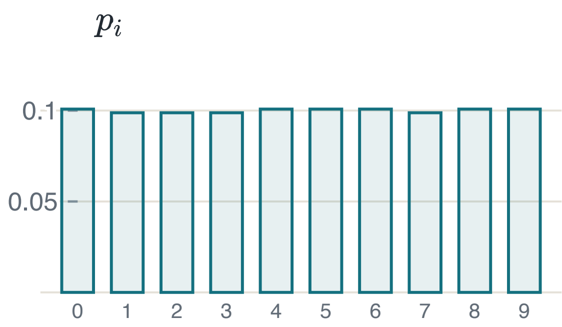
Whenever two nodes have the same probability, the algorithm may merge either, and different choices give different codebooks with the same average length. Your codewords will not match the ones drawn above and need not. What must match is $\bar{L}$, because Huffman is optimal and the optimal average length is unique even when the code is not. If two groups get different average lengths for the same statistics, one of them has an error. If they get different codebooks, neither of them does. Section 6.7 develops the tie-breaking rule that also minimises the variance of the codeword length.

The second part encodes one thousand symbols from the same distribution. Four-bit fixed-length coding uses 4000 bits. Huffman coding uses 2621 bits, so the measured compression is 34.5%. The average length predicts $1 - \bar{L}/4 = 34.1\%$. These values need not match exactly because the finite sequence is one random sample. A large difference can indicate an incorrect codebook.

The third part asks what the source statistics themselves are worth. Ten counts are drawn from a binomial trial with 50 attempts and success probability ρ, then normalised into a distribution. At $\rho = 0.01$ almost every attempt fails, so most counts come out zero and the surviving symbols are few. At $\rho = 0.99$ almost every attempt succeeds, so all ten counts come out near 50 and the distribution is nearly flat.



$\rho = 0.01$. Five symbols survive and the rest have probability zero. $H = 2.322$ bits, $\bar{L} = 2.400$ bits.



$\rho = 0.99$. All ten symbols occur about equally often. $H = 3.322$ bits, $\bar{L} = 3.395$ bits.

SOURCE	H	$\bar{L}$	η	COMPRESSION AGAINST 4 BITS
The digit source	2.550	2.636	96.7%	34.1%
$\rho = 0.01$	2.322	2.400	96.7%	40.0%
$\rho = 0.99$	3.322	3.395	97.8%	15.1%

EFFICIENCY AND COMPRESSION

The coding efficiency changes from 96.7% to 97.8%, while compression decreases from 40% to 15%. Efficiency measures the gap between the code length and the source entropy. Compression compares the code length with a four-bit fixed-length code. At $\rho = 0.99$, the source is nearly uniform and has entropy 3.322 bits. Therefore, little lossless compression remains possible.